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Back-end-of-line semiconductor device structure providing a not-gate logic function and methods of forming the same

Abstract

A semiconductor device structure providing a NOT gate logic function includes a layer stack including a pair of semiconductor layers having opposite conductivity-types, and a dielectric isolation layer disposed therebetween. First and second electrodes are located on a first side of the layer stack, where the first electrode contacts a first side surface of a first semiconductor layer and a second electrode contacts a first side surface a second semiconductor layer. A third electrode located on a second side of the layer stack contacts a second side surface of the first semiconductor layer and a second side surface of the second semiconductor layer. A gate dielectric layer is located over two side surfaces of the layer stack. A pair of gate electrodes located on either side of the layer stack contacts the gate dielectric layer. The semiconductor device structure may be fabricated using a BEOL process using metal-oxide semiconductor materials.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2021/0328056	12/2020	Baek et al.	N/A	N/A
2023/0058626	12/2022	Chuang et al.	N/A	N/A

OTHER PUBLICATIONS

Jianq Chyun Intellectual Property Office; Taiwanese Patent Application No. 112125333; First Office Action mailed Jan. 11, 2024; 8 pages. cited by applicant

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Background/Summary

BACKGROUND

(1) The semiconductor industry has grown due to continuous improvements in integration density of various electronic components (e.g., transistors, diodes, resistors, inductors, capacitors, etc.). For the most part, these improvements in integration density have come from successive reductions in minimum feature size, which allow more components to be integrated into a given area. In this regard, individual transistors, interconnects, and related structures have become increasingly smaller and there is an ongoing need to develop new materials, processes, and designs of semiconductor devices and interconnects to allow further progress.

(2) Transistors made of oxide semiconductors are an attractive option for back-end-of-line (BEOL) integration since such transistors may be processed at low temperatures and thus, may not damage previously fabricated devices. For example, the fabrication conditions and techniques may not damage previously fabricated front-end-of-line (FEOL) and middle end-of-line (MEOL) devices. Circuits based on oxide semiconductor-based transistor devices may further include other components that may be fabricated in a BEOL process, such as capacitors, inductors, resistors, and integrated passive devices.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of this disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. **1** is a vertical cross-sectional view of a first structure prior to formation of a semiconductor device structure providing a NOT-gate logic function according to various embodiments of the present disclosure.

(3) FIG. **2**A is a three-dimensional perspective view of a semiconductor device structure according to various embodiments of the present disclosure.

(4) FIG. **2**B is a further three-dimensional perspective view of the semiconductor device structure of FIG. **2**A.

(5) FIG. **2**C is a schematic equivalent circuit diagram describing the operations of the semiconductor device structure of FIG. **2**A according to various embodiments of the present disclosure.

(6) FIG. **2**D is a further three-dimensional perspective view of the semiconductor device structure of FIG. **2**A showing various dimensions of components of the semiconductor device structure according to various embodiments of the present disclosure.

(7) FIG. **3**A is a top view of an intermediate structure during a process of fabricating a semiconductor device structure as shown in FIGS. **2**A and **2**B according to various embodiments of the present disclosure.

(8) FIG. **3**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **3**A.

(9) FIG. **4**A is a top view of the intermediate structure illustrating a continuous first conductivity-type semiconductor layer formed over a first dielectric material layer according to various embodiments of the present disclosure.

(10) FIG. **4**B is a vertical cross-sectional view of the intermediate structure taken along line A-A′ in FIG. **4**B.

(11) FIG. **5**A is a top view of the intermediate structure illustrating a patterned mask formed over the continuous first conductivity-type semiconductor layer according to various embodiments of the present disclosure.

(12) FIG. **5**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **5**A.

(13) FIG. **6**A is a top view of the intermediate structure illustrating a discrete first conductivity-type semiconductor layer over the first dielectric material layer according to various embodiments of the present disclosure.

(14) FIG. **6**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **6**A.

(15) FIG. **7**A is a top view of the intermediate structure illustrating a second dielectric material layer formed over the first dielectric material layer and laterally surrounding the first conductivity-type semiconductor layer according to various embodiments of the present disclosure.

(16) FIG. **7**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **7**A.

(17) FIG. **8**A is a top view of the intermediate structure illustrating a patterned mask formed over the second dielectric material layer and the first conductivity-type semiconductor layer according to various embodiments of the present disclosure.

(18) FIG. **8**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **8**A.

(19) FIG. **9**A is a top view of the intermediate structure illustrating an opening formed through the

second dielectric material layer and exposing the upper surface of the first dielectric material layer and the upper surface of a conductive via according to various embodiments of the present disclosure.

(20) FIG. **9**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **9**A.

(21) FIG. **10**A is a top view of the intermediate structure illustrating a first electrode formed within the opening in the second dielectric material layer according to various embodiments of the present disclosure.

(22) FIG. **10**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **10**A.

(23) FIG. **11**A is a top view of the intermediate structure illustrating a dielectric isolation layer formed over the first conductivity-type semiconductor layer, the first electrode and the second dielectric material layer according to various embodiments of the present disclosure.

(24) FIG. **11**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **11**A.

(25) FIG. **12**A is a top view of the intermediate structure illustrating a continuous second conductivity-type semiconductor layer formed over the dielectric isolation layer according to various embodiments of the present disclosure.

(26) FIG. **12**B is a vertical cross-sectional view of the intermediate structure taken along line A-A′ in FIG. **12**B.

(27) FIG. **13**A is a top view of the intermediate structure illustrating a patterned mask formed over the continuous second conductivity-type semiconductor layer according to various embodiments of the present disclosure.

(28) FIG. **13**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **13**A.

(29) FIG. **14**A is a top view of the intermediate structure illustrating a discrete second conductivity-type semiconductor layer over the dielectric isolation layer according to various embodiments of the present disclosure.

(30) FIG. **14**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **14**A.

(31) FIG. **15**A is a top view of the intermediate structure illustrating a third dielectric material layer formed over the second dielectric material layer and laterally surrounding the second conductivity-type semiconductor layer according to various embodiments of the present disclosure.

(32) FIG. **15**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **15**A.

(33) FIG. **16**A is a top view of the intermediate structure illustrating a patterned mask formed over the third dielectric material layer and the second conductivity-type semiconductor layer according to various embodiments of the present disclosure.

(34) FIG. **16**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **16**A.

(35) FIG. **17**A is a top view of the intermediate structure illustrating an opening formed through the third dielectric material layer and exposing the upper surface of the dielectric isolation layer according to various embodiments of the present disclosure.

(36) FIG. **17**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **17**A.

(37) FIG. **18**A is a top view of the intermediate structure illustrating a second electrode formed within the opening in the third dielectric material layer according to various embodiments of the present disclosure.

(38) FIG. **18**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **18**A.

(39) FIG. **19**A is a top view of the intermediate structure illustrating a patterned mask formed over the third dielectric material layer, the second conductivity-type semiconductor layer, and the second electrode according to various embodiments of the present disclosure.
(40) FIG. **19**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **19**A.
(41) FIG. **20**A is a top view of the intermediate structure illustrating an opening formed through the third dielectric material layer, the dielectric isolation layer, and the second dielectric material layer and exposing the upper surface of the first dielectric layer according to various embodiments of the present disclosure.
(42) FIG. **20**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **20**A.
(43) FIG. **21**A is a top view of the intermediate structure illustrating a third electrode formed within the opening in the third dielectric material layer, the dielectric isolation layer, and the second dielectric material layer according to various embodiments of the present disclosure.
(44) FIG. **21**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **21**B.
(45) FIG. **22**A is a top view of the intermediate structure illustrating a patterned mask formed over the third dielectric material layer, the second conductivity-type semiconductor layer, the second electrode, and the third electrode according to various embodiments of the present disclosure.
(46) FIG. **22**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **22**A.
(47) FIG. **22**C is a vertical cross-section view of the intermediate structure taken along line B-B′ in FIG. **22**A.
(48) FIG. **23**A is a top view of the intermediate structure illustrating a pair of trench openings formed through the third dielectric material layer, the dielectric isolation layer, and the second dielectric material layer and exposing the upper surface of the first dielectric layer according to various embodiments of the present disclosure.
(49) FIG. **23**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **23**A.
(50) FIG. **23**C is a vertical cross-section view of the exemplary intermediate structure taken along line B-B′ in FIG. **23**A.
(51) FIG. **24**A is a top view of the intermediate structure illustrating a gate dielectric layer formed over the third dielectric material layer, the second conductivity-type semiconductor layer, the second electrode, and the third electrode and over the sidewalls and bottom surface of the trench openings according to various embodiments of the present disclosure.
(52) FIG. **24**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **24**A.
(53) FIG. **24**C is a vertical cross-section view of the intermediate structure taken along line B-B′ in FIG. **24**A.
(54) FIG. **25**A is a top view of the intermediate structure illustrating a dielectric fill layer formed over the gate dielectric layer and filling a remaining volume of the trench openings and a patterned mask formed over the dielectric fill layer according to various embodiments of the present disclosure.
(55) FIG. **25**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **25**A.
(56) FIG. **25**C is a vertical cross-section view of the intermediate structure taken along line B-B′ in FIG. **25**A.
(57) FIG. **26**A is a top view of the intermediate structure illustrating a pair of trench openings formed through the dielectric fill layer and portions of the gate dielectric layer and exposing the upper surface of the first dielectric layer according to various embodiments of the present

disclosure.
(58) FIG. **26**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **26**A.
(59) FIG. **26**C is a vertical cross-section view of the intermediate structure taken along line B-B′ in FIG. **26**A.
(60) FIG. **27**A is a top view of a semiconductor device structure including a pair of gate electrodes extending on opposite sides of a layer stack according to various embodiments of the present disclosure.
(61) FIG. **27**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **27**A.
(62) FIG. **27**C is a vertical cross-section view of the intermediate structure taken along line B-B′ in FIG. **27**A.
(63) FIG. **27**D is a vertical cross-section view of the intermediate structure taken along line C-C′ in FIG. **27**A.
(64) FIG. **27**E is a vertical cross-section view of the intermediate structure taken along line D-D′ in FIG. **27**A.
(65) FIG. **28**A is a top view of a semiconductor device structure including a fourth dielectric material layer formed over the gate dielectric layer and the gate electrodes and a plurality of conductive vias located within the fourth dielectric material layer according to various embodiments of the present disclosure.
(66) FIG. **28**B is a vertical cross-section view of the semiconductor device structure **100** of taken along line A-A′ in FIG. **28**A.
(67) FIG. **28**C is a vertical cross-section view of the semiconductor device structure **100** of taken along line B-B′ in FIG. **28**A.
(68) FIG. **29** is a flow diagram illustrating a method for fabricating a semiconductor device structure according to various embodiments of the present disclosure.

DETAILED DESCRIPTION

(69) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify this disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, this disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
(70) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. Unless explicitly stated otherwise, each element having the same reference numeral is presumed to have the same material composition and to have a thickness within a same thickness range.
(71) Various embodiments of this disclosure provide semiconductor device structures and methods that may be advantageous in terms of manufacturing flexibility, improved integration density, and increased computing power of semiconductor integrated circuit (IC) dies. In this regard, an

embodiment semiconductor device structure is provided that may be formed in a BEOL process and may be incorporated with other BEOL circuit components such as capacitors, inductors, resistors, and integrated passive devices. As such, the disclosed semiconductor device structure may include materials that may be processed at low temperatures and thus, may not damage previously fabricated devices (e.g., FEOL and MEOL devices).

(72) In various embodiments, the semiconductor device structure may be configured as a logic NOT gate or inverter circuit such that applying an input signal to the semiconductor device structure having a first logic state will produce an output signal from the semiconductor device structure having the complementary logic state. In this regard, providing a high voltage input signal to the semiconductor device structure may result in a low voltage output signal from the semiconductor device structure, and providing a low voltage input signal to the semiconductor device structure may result in a high voltage output signal from the semiconductor device structure.

(73) In various embodiments, the semiconductor device structure may include a layer stack including a first conductivity-type semiconductor layer, a dielectric isolation layer over the first conductivity-type semiconductor layer, and a second conductivity-type semiconductor layer over the dielectric isolation layer. In various embodiments, the first conductivity-type semiconductor layer and/or the second conductivity-type semiconductor layer may include a metal oxide semiconductor material. A first electrode and a second electrode may be located on a first side of the layer stack, where the first electrode may contact a first side surface of the first conductivity-type semiconductor layer and the second electrode may contact a first side surface of the second conductivity-type semiconductor layer. The dielectric isolation layer may extend between the first electrode and the second electrode. A third electrode may be located on a second side of the layer stack, opposite the first side, and may contact a second side surface of the first conductivity-type semiconductor layer, a side surface of the dielectric isolation layer, and a second side surface of the second conductivity-type semiconductor layer. A gate dielectric layer may surround the layer stack over two side surfaces and an upper surface of the layer stack over at least a portion of a length of the layer stack between the first and second electrodes on one end of the layer stack and the third electrode at the other end of the layer stack. A pair of gate electrodes may be located on either side of the layer stack, where the gate dielectric layer may be located between each of the gate electrodes and the respective opposite side surfaces of the layer stack over at least a portion of the length of the layer stack between first and second electrodes on one end of the layer stack and the third electrode at the other end of the layer stack.

(74) In various embodiments, semiconductor device structures providing a NOT gate logic functions may be fabricated using a BEOL process using thin-film metal oxide semiconductor materials. Accordingly, this may enable computing operations to be performed, at least in part, within one or more interconnect levels of a semiconductor integrated circuit (IC) die, which may enable improved integration density and/or increased processing power of the semiconductor IC die.

(75) FIG. **1** is a vertical cross-sectional view of a first structure prior to formation of a semiconductor device structure providing a NOT-gate logic function according to various embodiments of the present disclosure. The first structure includes a substrate **8**, which may be a semiconductor substrate such as a commercially available silicon substrate. The substrate **8** may include a semiconductor material layer **9** at least at an upper portion thereof. The semiconductor material layer **9** may be a surface portion of a bulk semiconductor substrate, or may be a top semiconductor layer of a semiconductor-on-insulator (SOI) substrate. In one embodiment, the semiconductor material layer **9** includes a single crystalline semiconductor material such as single crystalline silicon. In one embodiment, the substrate **8** may include a single crystalline silicon substrate including a single crystalline silicon material.

(76) Shallow trench isolation structures **720** including a dielectric material such as silicon oxide may be formed in an upper portion of the semiconductor material layer **9**. Suitable doped

semiconductor wells, such as p-type wells and n-type wells, may be formed within each area that is laterally enclosed by a portion of the shallow trench isolation structures **720**. Field effect transistors **701** may be formed over the top surface of the semiconductor material layer **9**. For example, each field effect transistor **701** may include a source electrode **732**, a drain electrode **738**, a semiconductor channel **735** that includes a surface portion of the substrate **8** extending between the source electrode **732** and the drain electrode **738**, and a gate structure **750**. The semiconductor channel **735** may include a single crystalline semiconductor material. Each gate structure **750** may include a gate dielectric layer **752**, a gate electrode **754**, a gate cap dielectric **758**, and a dielectric gate spacer **756**. A source-side metal-semiconductor alloy region **742** may be formed on each source electrode **732**, and a drain-side metal-semiconductor alloy region **748** may be formed on each drain electrode **738**. The devices formed on the top surface of the semiconductor material layer **9** may include complementary metal-oxide-semiconductor (CMOS) transistors and optionally additional semiconductor devices (such as resistors, diodes, capacitor structures, etc.), and are collectively referred to as CMOS circuitry **700**.

(77) One or more of the field effect transistors **701** in the CMOS circuitry **700** may include a semiconductor channel **735** that contains a portion of the semiconductor material layer **9** in the substrate **8**. If the semiconductor material layer **9** includes a single crystalline semiconductor material such as single crystalline silicon, the semiconductor channel **735** of each field effect transistor **701** in the CMOS circuitry **700** may include a single crystalline semiconductor channel such as a single crystalline silicon channel. In one embodiment, a subset of the field effect transistors **701** in the CMOS circuitry **700** may include a respective node that is subsequently electrically connected to a node of a NOT gate semiconductor device structure to be subsequently formed.

(78) In one embodiment, the substrate **8** may include a single crystalline silicon substrate, and the field effect transistors **701** may include a respective portion of the single crystalline silicon substrate as a semiconducting channel. As used herein, a “semiconducting” element refers to an element having electrical conductivity in the range from 1.0×10.sup.−6 S/cm to 1.0×10.sup.5 S/cm. As used herein, a “semiconductor material” refers to a material having electrical conductivity in the range from 1.0×10.sup.−6 S/cm to 1.0×10.sup.5 S/cm in the absence of electrical dopants therein, and is capable of producing a doped material having electrical conductivity in a range from 1.0 S/cm to 1.0×10.sup.5 S/cm upon suitable doping with an electrical dopant.

(79) Various metal interconnect structures formed within dielectric material layers may be subsequently formed over the substrate **8** and the semiconductor devices **701** thereupon (such as field effect transistors). In an illustrative example, the dielectric material layers may include, for example, a first dielectric material layer **601** that may be a layer that surrounds the contact structure connected to the source and drains (sometimes referred to as a contact-level dielectric material layer **601**), a first interconnect-level dielectric material layer **610**, a second interconnect-level dielectric material layer **620**, a third interconnect-level dielectric material layer **630**, and a fourth interconnect-level dielectric material layer **640**. The metal interconnect structures may include device contact via structures **612** formed in the first dielectric material layer **601** and contact a respective component of the CMOS circuitry **700**, first metal line structures **618** formed in the first interconnect-level dielectric material layer **610**, first metal via structures **622** formed in a lower portion of the second interconnect-level dielectric material layer **620**, second metal line structures **628** formed in an upper portion of the second interconnect-level dielectric material layer **620**, second metal via structures **632** formed in a lower portion of the third interconnect-level dielectric material layer **630**, third metal line structures **638** formed in an upper portion of the third interconnect-level dielectric material layer **630**, third metal via structures **642** formed in a lower portion of the fourth interconnect-level dielectric material layer **640**, and fourth metal line structures **648** formed in an upper portion of the fourth interconnect-level dielectric material layer **640**. While the present disclosure is described using an embodiment in which four levels metal line

structures are formed in dielectric material layers, embodiments are expressly contemplated herein in which a lesser or greater number of levels of metal line structures are formed in dielectric material layers.

(80) Each of the dielectric material layers (**601**, **610**, **620**, **630**, **640**) may include a dielectric material such as undoped silicate glass, a doped silicate glass, organosilicate glass, amorphous fluorinated carbon, porous variants thereof, or combinations thereof. Each of the metal interconnect structures (**612**, **618**, **622**, **628**, **631**, **638**, **642**, **648**) may include at least one conductive material, which may be a combination of a metallic liner (such as a metallic nitride or a metallic carbide) and a metallic fill material. Each metallic liner may include TIN, TaN, WN, TiC, TaC, and WC, and each metallic fill material portion may include W, Cu, Al, Co, Ru, Mo, Ta, Ti, alloys thereof, and/or combinations thereof. Other suitable metallic liner and metallic fill materials within the contemplated scope of disclosure may also be used. In one embodiment, the first metal via structures **622** and the second metal line structures **628** may be formed as integrated line and via structures by a dual damascene process. Generally, any contiguous set of a metal line structure (**628**, **638**, **648**) and at least one underlying metal via structure (**622**, **632**, **642**) may be formed as an integrated line and via structure.

(81) Generally, semiconductor devices **701** may be formed on a substrate **8**, and metal interconnect structures (**612**, **618**, **622**, **628**, **631**, **638**, **642**, **648**) and dielectric material layers (**601**, **610**, **620**, **630**, **640**) over the semiconductor devices **701**. The metal interconnect structures (**612**, **618**, **622**, **628**, **631**, **638**, **642**, **648**) may be formed in the dielectric material layers (**601**, **610**, **620**, **630**, **640**), and may be electrically connected to the semiconductor devices.

(82) Referring again to FIG. **1**, a first dielectric material layer **102** may be formed over the metal interconnect structures (**612**, **618**, **622**, **628**, **631**, **638**, **642**, **648**) and dielectric material layers (**601**, **610**, **620**, **630**, **640**). The first dielectric material layer **102** may include a suitable dielectric material, such as silicon oxide, undoped silicate glass, a doped silicate glass, organosilicate glass, amorphous fluorinated carbon, porous variants thereof, or combinations thereof. Other suitable dielectric materials are within the contemplated scope of disclosure. The first dielectric material layer **102** may be deposited using any suitable deposition process, such a chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), high density plasma CVD (HDPCVD), metalorganic CVD (MOCVD), plasma enhanced CVD (PECVD), sputtering, laser ablation, or the like. In some embodiments, one or more above-described metal interconnect structures, such as integrated line and via structures, may be formed within the first dielectric material layer **102** and may be coupled to metal interconnect structures (**612**, **618**, **622**, **628**, **631**, **638**, **642**, **648**) located within the underlying dielectric material layers (**601**, **610**, **620**, **630**, **640**). The first dielectric material layer **102** may include a planar upper surface **122**.

(83) FIG. **2**A is a three-dimensional perspective view of a semiconductor device structure **100**, and FIG. **2**B is a further three-dimensional perspective view of the semiconductor device structure **100** of FIG. **2**A that has been rotated, according to various embodiments. FIG. **2**C is a schematic equivalent circuit diagram **100***c* describing the operations of the semiconductor device structure **100** of FIGS. **2**A and **2**B, according to various embodiments. A semiconductor device structure **100** as shown in FIGS. **2**A and **2**B may be formed via a BEOL process. In some embodiments, the semiconductor device structure **100** may be formed on and/or in a dielectric material layer, such as the first dielectric material layer **102** described above with reference to FIG. **1**. The semiconductor device structure **100** may be embedded in an insulating matrix that may include one or more dielectric material layers (not shown in FIGS. **2**A and **2**B for clarity).

(84) Referring to FIGS. **2**A and **2**B, the semiconductor device structure **100** may include a first end **101** and a second end **104** that is opposite the first end **101** along a first horizontal direction hd**1**. A first electrode **113** and a second electrode **115** may be located at the first end **101** of the semiconductor device structure **100**. The second electrode **115** may be located vertically above the first electrode **113**. A layer of dielectric material **111** (which may also be referred to as a dielectric

isolation layer **111**) may be located between first electrode **113** and the second electrode **115**. A third electrode **117** may be located at the second end **104** of the semiconductor device structure **100**.
(85) A layer stack **170** may extend along the first horizontal direction hd**1** between the first electrode **113** and second electrode **115** on one end (e.g., **101**) of the layer stack **170** and the third electrode **117** on the opposite end (e.g., **104**) of the layer stack **170**. The layer stack **170** may include a first conductivity-type semiconductor layer **107**, the dielectric isolation layer **111** over the first conductivity-type semiconductor layer **107**, and a second conductivity-type semiconductor layer **109** over the dielectric isolation layer **111**. For example, when the first conductivity-type semiconductor layer **107** may be p-type, the second conductivity-type semiconductor layer **109** may be n-type, and vice versa. In one non-limiting embodiment, the first conductivity-type semiconductor layer **107** may be composed of a p-type semiconductor material and the second conductivity-type semiconductor layer **109** may composed of an n-type semiconductor material. In some embodiments, the first conductivity-type semiconductor layer **107** and the second conductivity-type semiconductor layer **109** may be composed of oxide semiconductor materials, as described in further detail below.
(86) The layer stack **170** may include a gate dielectric layer **103** that may surround the layer stack **170** over two side surfaces and an upper surface of the layer stack **170** over at least a portion of the layer stack **170** between the third electrode **117** on one end (e.g., **104**) of the layer stack **170** and the first electrode **113** and the second electrode **115** on the other end (e.g., **101**) of the layer stack **170**. In other words, the gate dielectric layer **103** may extend over and contact an upper surface and two opposite side surfaces of the second conductivity-type semiconductor layer **109**, two opposite side surfaces of the dielectric isolation layer **111**, and two opposite side surfaces of the first conductivity-type semiconductor layer **107**. In some embodiments, the gate dielectric layer **103** may also contact a bottom surface of the first conductivity-type semiconductor layer **107**. A pair of gate electrodes **105***a* and **105***b* may be located on either side of the layer stack **170**, where the gate dielectric layer **103** may be located between each of the gate electrodes **105***a* and **105***b* and the respective opposite side surfaces of the layer stack **170** over at least a portion of the length of the layer stack **170**. Although FIGS. **2**A and **2**B illustrate the gate dielectric layer **103** and the gate electrodes **105***a* and **105***b* extending along a portion of the length of the layer stack **170**, it will be understood that in other embodiments, the gate dielectric layer **103** and the gate electrodes **105***a* and **105***b* may extend over the entire or substantially entire length of the layer stack **170**.
(87) A first end of the first conductivity-type semiconductor layer **107** may contact a side surface of the first electrode **113** and a second end of the first conductivity-type semiconductor layer **107** may contact a side surface of the third electrode **117**. A first end of the second conductivity-type semiconductor layer **109** may contact a side surface of the second electrode **115** and a second end of the second conductivity-type semiconductor layer **109** may contact the side surface of the third electrode **117**. As schematically illustrated in FIGS. **2**B and **2**C, the first electrode **113** may be electrically connected to a voltage supply (e.g., that may be held at a source voltage VDD). The second electrode **115** may be electrically connected to a ground voltage terminal (e.g., that may be held at a ground (GND) voltage). The pair of gate electrodes **105***a* and **105***b* may be electrically connected to a common input voltage terminal (e.g., Vin). The third electrode **117** may be electrically connected to an output signal terminal (e.g., Vout).
(88) The semiconductor device structure **100** as shown in FIGS. **2**A and **2**B may be configured as a NOT gate circuit (which may also be referred to as an inverter circuit). A NOT gate circuit is a logic gate circuit that inverts (i.e., outputs the complementary logic state) of a given input signal. For example, when a low voltage signal (e.g., representing a logic state of “0” or “1”) is provided at the input terminal (Vin) of the semiconductor device structure **100**, the output signal terminal (Vout) of the semiconductor device structure **100** outputs a high voltage signal (e.g., representing the complementary logic state of the input signal, such that when the input low voltage signal

represents a logic state “0,” the output high voltage signal represents a logic state “1,” and vice versa). Similarly, when a high voltage signal is provided at the input terminal (Vin), a low voltage signal is output at the output signal terminal (Vout).
(89) FIG. **2**C is a schematic equivalent circuit diagram **100***c* that describes the operation of the semiconductor device structure **100** as shown in FIGS. **2**A and **2**B. In particular, the first conductivity-type semiconductor layer **107** may be configured as a channel layer of a first transistor **201**, which may be a p-channel metal oxide semiconductor field effect transistor (MOSFET). Thus, the first transistor **201** may include the first conductivity-type semiconductor layer **107** (which may function as a p-channel layer), the first electrode **113** (which may function as a source electrode), the third electrode **117** (which may function as a drain electrode), the gate dielectric layer **103** and the pair of gate electrodes **105***a* and **105***b* on either side of the p-channel layer **107**. Similarly, the second conductivity-type semiconductor layer **109** may be configured as a channel layer of a second transistor **203**, which may be an n-channel metal oxide semiconductor field effect transistor (MOSFET). Thus, the second transistor **203** may include the second conductivity-type semiconductor layer **109** (which may function as an n-channel layer), the second electrode **115** (which may function as a source electrode), the third electrode **117** (which may function as a drain electrode), the gate dielectric layer **103** and the pair of gate electrodes **105***a* and **105***b* on either side of the n-channel layer **109**. It will be understood that source/drain electrode(s) **113**, **115** and **117** may refer to a source or a drain electrode, individually or collectively, dependent upon the context.
(90) Referring again to FIG. **2**C, a low voltage placed on the input signal terminal Vin turns on the first transistor **201** and turns off the second transistor **203**. Since the source of the first transistor **201** (i.e., the first electrode **113**) is connected to the voltage supply (VDD) that has a high voltage, the output voltage Vout (i.e., the voltage at the third electrode **117**) will have a high voltage. Similarly, a high voltage placed on the input signal terminal Vin turns on the second transistor **203** and turns off the first transistor **201**. Since the source of the second transistor **203** (i.e., the second electrode **115**) is connected to a ground voltage terminal, the output voltage Vout (i.e., the voltage at the third electrode **117**) will have a low voltage. In this way, a high input signal is converted to a low output signal and a low input signal is converted to a high input signal. As such, the semiconductor device structure **100** is configured as a NOT gate or inverter circuit.
(91) In some embodiments, the semiconductor device structure **100** of FIGS. **2**A and **2**B may be formed over an interconnect-level dielectric layer having a planar horizontal surface. For example, the semiconductor device structure **100** may be formed over the first dielectric material layer **102** (e.g., see FIG. **1**), over any of layers **601**, **610**, **620**, **630** and **640** in FIG. **1**, or over one or more additional interconnect-level dielectric layers formed over the first dielectric material layer **102**. As shown in FIGS. **2**A and **2**B, the layer stack **170** including the first conductivity-type semiconductor layer **107**, the dielectric isolation layer **111** and the second conductivity-type semiconductor layer **109** may have a horizontal orientation such that the interfacing surfaces of the first conductivity-type semiconductor layer **107**, the dielectric isolation layer **111** and the second conductivity-type semiconductor layer **109** may be substantially parallel to the planar horizontal surface of the interconnect-level dielectric layer (e.g., see FIG. **1** in which the first dielectric material layer **102** has a horizontal upper surface **122**). However, it will be understood that other orientations of the semiconductor device structure **100** are within the contemplated scope of disclosure. For example, the semiconductor device structure **100** may be oriented such that the interfacing surfaces of the gate electrodes **105***a*, **105***b*, the gate dielectric layer **103**, and the respective opposite side surfaces of the layer stack **170** may be substantially parallel to the planar horizontal surface of the interconnect-level dielectric layer.
(92) As discussed above, the interconnect-level dielectric layer on which the semiconductor device structure **100** is formed may include one or more electrical interconnect structures. In this regard, one or more of the first electrode **113**, the second electrode **115**, the third electrode **117** and the gate electrodes **105***a* and **105***b* may be electrically connected to the one or more electrical interconnect

structures (e.g., **612**, **618**, **622**, **628**, **632**, **638**, **642**, **648**) formed in one or more dielectric material layers (**601**, **610**, **620**, **630**, **640**) underlying the semiconductor device structure **100**. In other embodiments, one or more of the first electrode **113**, the second electrode **115**, the third electrode **117** and the gate electrodes **105***a* and **105***b* may be electrically connected to one or more electrical interconnect structures to be subsequently formed over the semiconductor device structure **100**.
(93) In one or more embodiments, one or both of the first conductivity-type semiconductor layer **107** and the second conductivity-type semiconductor layer **109** may include metal-oxide semiconductors. For example, suitable metal-oxide semiconductor materials for a p-type semiconductor layer **107** or **109** may include one or more of NiO, SnO, and Cu.sub.2O. Other suitable p-type semiconductor materials are within the contemplated scope of disclosure. Suitable metal-oxide semiconductor materials for an n-type semiconductor layer **109** or **107** may include one or more of InZnO (IZO), InSnO (indium tin oxide (ITO), In.sub.2O.sub.3, Ga.sub.2O.sub.3, InGaZnO (IGZO), Al.sub.2O.sub.5Zn.sub.2, ZnO, InGaO, InWO, ZnO, TiOx, aluminum-doped zinc oxide (AZO), including various combinations and alloys thereof. Other suitable n-type semiconductor materials, such as amorphous silicon, III-V materials, and the like, including combinations (e.g., stacked layers) and/or alloys thereof, may also be utilized. In some embodiments, the n-type semiconductor layer **109** or **107** may have a composition given by In.sub.x Ga.sub.y Zn.sub.z MO, wherein $0<x<1$; $0\leq y\leq 1$; $0\leq z\leq 1$; and M is one of Ti, Al, Ag, Ce, and Sn. In other embodiments, the n-type semiconductor layer **109** may include an alloy of oxygen, a group-III element, and a group-V element. In other embodiments, the one or more of the n-type semiconductor layer **109** or **107** and the p-type semiconductor layer **107** or **109** may be formed of a metal-oxide semiconductor having a multi-layer structure.
(94) In some embodiments, the dielectric isolation layer **111** may include one or more of AlOx, SiO.sub.2, SiNx, or other interconnect-level dielectric materials, as described above. The gate dielectric layer **103** may include one or more of silicon oxide, aluminum oxide, hafnium oxide, hafnium lanthanum oxide, hafnium silicon oxide, hafnium tantalum oxide, hafnium titanium oxide, hafnium zirconium oxide, zirconium oxide, titanium oxide, tantalum oxide, hafnium dioxide-alumina, etc. In some embodiments, the gate dielectric layer **103** may include a high-k dielectric material. Other suitable materials for the dielectric isolation layer **111** and/or the gate dielectric layer **103** are within the contemplated scope of disclosure.
(95) As described in greater detail below, one or more of the first electrode **113**, the second electrode **115**, the third electrode **117** and the gate electrodes **105***a* and **105***b* may include one or more of TiN, W, WN, WCN, Co, PdCo, Mo, Cu, TaN, Ti, Al, etc. Other suitable conductor materials may be within the contemplated scope of disclosure.
(96) FIG. **2**D is a further three-dimensional perspective view showing various dimensions of components of the semiconductor device structure **100** of FIGS. **2**A and **2**B, according to various embodiments. As described above, the first conductivity-type semiconductor layer **107** may be configured as a p-channel layer of a first transistor **201** and the second conductivity type semiconductor layer **109** may be configured as an n-channel layer of second transistor **203** (e.g., see FIG. **2**C). As such, when the respective transistors (**201**, **203**) are activated, current may flow as indicated by the dashed arrows (**220***a*, **220***b*) in FIG. **2**D. In this regard, when the first transistor **201** is activated (e.g., by applying a low or zero bias to the gate electrodes **105***a* and **105***b*) positive charge carriers (i.e., “holes”) may flow from the first electrode **113** to the third electrode **117** giving rise to a first current **220***a*. Similarly, when the second transistor **203** is activated (e.g., by applying a high bias to the gate electrodes **105***a* and **105***b*) negative charge carriers (i.e., electrons) may flow from the third electrode **117** to the second **115** but, since the current carried by a negative charge is opposite to its motion, the charge motion in the n-channel **109** gives rise to a second current **220***b*, which is in the same direction as first current **220***a* that flows in the p-channel **107**.
(97) Each of the p-type semiconductor layer **107** and the n-type semiconductor layer **109** may have a respective channel length **222** (i.e., along the first horizontal direction hd**1**) and a respective

channel width **224** (i.e., along a second horizontal direction hd**2** that is perpendicular to the first horizontal direction hd**1**). The channel length **222** may have a value greater than 15 nm in various embodiments. An increased value of the channel length **222** may mitigate short channel effects. However, increasing the channel length **222** may result in reduced driving current and a greater size of the semiconductor circuit **100**. Thus, it may be possible to optimize the channel length **222** to determine a value sufficiently large to avoid short channel effects while also keeping the size of the semiconductor device structure **100** as small as possible. The channel width **224** may have a value that is greater than 5 nm and less than 500 nm according to various embodiments.

(98) Each of the first electrode **113**, the second electrode **115** and the third electrode **117** may have a width along the second horizontal direction hd**2** that is approximately equal to the channel width **224**. Each of the first electrode **113**, the second electrode **115** and the third electrode **117** may also have a length dimension **226** (along hd**1**) that is greater than 5 nm. The first electrode **113** and the second electrode **115** may have thickness dimensions **228** and **229** (along a vertical direction) between about 2 nm and about 50 nm. As shown in FIG. **2**D, the third electrode **117** may have a thickness dimension **231** that is greater than the combined thickness dimensions **228** and **229** of the first electrode **113** and the second electrode **115**. The thickness dimension **231** of the third electrode **117** may be approximately equal to the total thickness of the layer stack **170** including the first conductivity-type semiconductor layer **107**, the dielectric isolation layer **111** and the second conductivity-type semiconductor layer **109**. The dielectric isolation layer **111** may have a thickness **230** that is greater than 5 nm and less than 500 nm. The gate electrodes **105**_a_ and **105**_b_ may each have a gate length **232** that may be comparable to the channel length **222** in some embodiments. For example, in some embodiments the gate length **232** may be greater than about 10 nm. The gate electrodes **105**_a_ and **105**_b_ may each have a gate width **234** that is at least about 10 nm. The gate dielectric layer may have a thickness **238** that is greater than about 2 nm and less than about 20 nm. The first conductivity-type semiconductor layer **107** may have a thickness **240** that is between about 2 nm and about 50 nm. The second conductivity-type semiconductor layer **109** may have a thickness **241** that is between about 2 nm and about 50 nm. In some embodiments, the thickness **240** of the first conductivity-type semiconductor layer **107** may be approximately equal to the thickness **228** of the first electrode **113**, and the thickness **241** of the second conductivity-type semiconductor layer **109** may be approximately equal to the thickness **229** of the second electrode **115**.

(99) FIG. **3**A is a top view of an intermediate structure during a process of fabricating a semiconductor device structure **100** as shown in FIGS. **2**A and **2**B according to various embodiments of the present disclosure. FIG. **3**B is a vertical cross-section view of the exemplary intermediate structure taken along line A-A′ in FIG. **3**A. Referring to FIGS. **3**A and **3**B, the exemplary intermediate structure may include a first dielectric material layer **102** having a planar upper surface **122**. The first dielectric material layer **102** may be formed in a BEOL process as described above with reference to FIG. **1**. Accordingly, the first dielectric material layer **102** may be formed over a substrate **8** including a semiconductor material layer **9**, semiconductor devices **701**, metal interconnect structures (**612**, **618**, **622**, **628**, **631**, **638**, **642**, **648**) and dielectric material layers (**601**, **610**, **620**, **630**, **640**), as described above with reference to FIG. **1**.

(100) The first dielectric material layer **102** may be composed of a suitable dielectric material, such as undoped silicate glass, a doped silicate glass (e.g., deposited by decomposition of tetraethylorthosilicate (TEOS)), organosilicate glass, silicon oxynitride, or silicon carbide nitride. Other suitable dielectric materials are within the contemplated scope of disclosure. The first dielectric material layer **102** may be deposited by a conformal deposition process (e.g., chemical vapor deposition (CVD), atomic layer deposition (ALD), physical vapor deposition (PVD), plasma enhanced chemical vapor deposition (PECVD), etc.) or a self-planarizing deposition process (such as spin coating). The thickness of the first dielectric material layer **102** be in a range from approximately 15 nm to approximately 60 nm, such as from approximately 20 nm to approximately

40 nm, although smaller and larger thicknesses may also be used.

(101) Referring again to FIGS. **3**A and **3**B, metal interconnect features, such as metal line **120** and via **121** structures, may be formed within the first dielectric material layer **102** (e.g., via a single or dual damascene process). Upper surfaces **123** of each of the vias **121** may be coplanar with the upper surface **122** of the first dielectric material layer **102**.

(102) FIG. **4**A is a top view of the intermediate structure illustrating a continuous first conductivity-type semiconductor layer **107**L formed over the first dielectric material layer **102** according to various embodiments of the present disclosure. FIG. **4**B is a vertical cross-sectional view of the intermediate structure taken along line A-A′ in FIG. **4**B. Referring to FIGS. **4**A and **4**B, a continuous first conductivity-type semiconductor layer **107**L may be deposited over the upper surface **122** of the first dielectric material layer **102** and the upper surfaces **123** of each of the conductive vias **121** via a suitable deposition process, such as ALD, CVD, PECVD, PVD, etc. A thickness of the continuous first conductivity-type semiconductor layer **107**L may be in a range from approximately 2 nm to approximately 50 nm, although other embodiments may include smaller and larger thicknesses. The continuous first conductivity-type semiconductor layer **107**L may be composed of a suitable semiconductor material, such as a metal oxide semiconductor material. In some embodiments, the continuous first conductivity-type semiconductor layer **107**L may be a p-type semiconductor material including, but not limited to, one or more of NiO, SnO, and Cu.sub.2O. Other suitable p-type semiconductor materials are within the contemplated scope of disclosure. In some embodiments, following the deposition of the continuous first conductivity-type semiconductor layer **107**L, the intermediate structure may optionally be annealed. The optional annealing process may be performed at a temperature in a range from 200° C. to 400° C. using a rapid thermal annealing or furnace annealing process. The annealing may be performed in an environment of nitrogen, oxygen, or a mixture thereof.

(103) FIG. **5**A is a top view of the intermediate structure illustrating a patterned mask **125** formed over the continuous first conductivity-type semiconductor layer **107**L according to various embodiments of the present disclosure. FIG. **5**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **5**A. Referring to FIGS. **5**A and **5**B, the patterned mask **125** may be formed by depositing a layer of a photoresist material over the upper surface **124** of the continuous first conductivity-type semiconductor layer **107**L and patterning the photoresist material using lithographic techniques to form a patterned photoresist mask **125** as shown in FIGS. **5**A and **5**B. In some embodiments, the patterned mask **125** may include a strip-shaped portion having a length along the first horizontal direction hd**1** of at least about 15 nm and a width along the second horizontal direction hd**2** of at least about 5 nm. The patterned mask **125** may then be used as a mask to pattern the continuous first conductivity-type semiconductor layer **107**L, as described in greater detail with reference to FIGS. **6**A and **6**B, below.

(104) FIG. **6**A is a top view of the intermediate structure illustrating a discrete first conductivity-type semiconductor layer **107** over the first dielectric material layer **102** according to various embodiments of the present disclosure. FIG. **6**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **6**A. Referring to FIGS. **6**A and **6**B, an anisotropic etching process may be performed to remove portions of the continuous first conductivity-type semiconductor layer **107** that are not masked by the patterned mask **125**. The patterned mask **125** may then be removed by ashing or by dissolution with a solvent. The remaining portion of the continuous first conductivity-type semiconductor layer **107** may include a discrete first conductivity-type semiconductor layer **107** over the upper surface **122** of the first dielectric material layer **102**. In some embodiments, the first conductivity-type semiconductor layer **107** may have a length along the first horizontal direction hd**1** of at least about 15 nm and a width along the second horizontal direction hd**2** of at least about 5 nm.

(105) FIG. **7**A is a top view of the intermediate structure illustrating a second dielectric material layer **126** formed over the first dielectric material layer **102** and laterally surrounding the first

conductivity-type semiconductor layer **107** according to various embodiments of the present disclosure. FIG. **7**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **7**A. Referring to FIGS. **7**A and **7**B, the second dielectric material layer **126** may be deposited over the upper surface **122** of the first dielectric material layer **102**, the upper surfaces **123** of the conductive vias **121**, and over the upper surface **110** and side surfaces of the first conductivity-type semiconductor layer **107** via a suitable deposition method (e.g., CVD, ALD, PVD, PECVD, spin coating, etc.). The second dielectric material layer **126** may be composed of a suitable dielectric material as described above (e.g., undoped silicate glass, a doped silicate glass, organosilicate glass, silicon oxynitride, silicon carbide nitride, etc.). An optional planarization process, such as a chemical mechanical planarization (CMP) process, may be performed to remove excess portions of the second dielectric material layer **126** from over the upper surface **110** of the first conductivity-type semiconductor layer **107** and provide a planar upper surface **127** of the second dielectric material layer **126**. In the embodiment of FIGS. **7**A and **7**D, the planar upper surface **127** of the second dielectric material layer **126** is substantially coplanar with the upper surface **110** of the first conductivity-type semiconductor layer **107**. In other embodiments, the planar upper surface **127** of the second dielectric material layer **126** may extend over the upper surface **110** of the first conductivity-type semiconductor layer **107**.

(106) FIG. **8**A is a top view of the intermediate structure illustrating a patterned mask **128** formed over the second dielectric material layer **126** and the first conductivity-type semiconductor layer **107** according to various embodiments of the present disclosure. FIG. **8**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **8**A. Referring to FIGS. **8**A and **8**B, the patterned mask **128** may be formed by depositing a layer of a photoresist material over the upper surface **127** of the second dielectric material layer **126** and the upper surface **110** of the first conductivity-type semiconductor layer **107** and patterning the photoresist material using lithographic techniques to form a patterned photoresist mask **128** as shown in FIGS. **8**A and **8**B. The patterned mask **128** may include a strip-shaped opening **129** through the patterned mask **128**, where a portion of the second dielectric material layer **126** may be exposed at the bottom of the opening **129** in the patterned mask **128**. In some embodiments, the opening **129** may have a dimension along the second horizontal direction hd**2** that is at least about 5 nm and may be substantially equal to the width dimension of the first conductivity-type semiconductor layer **107** along the second horizontal direction hd**2**. The opening **129** may have a dimension along the first horizontal direction hd**1** of at least about 5 nm.

(107) FIG. **9**A is a top view of the intermediate structure illustrating an opening **130** formed through the second dielectric material layer **126** and exposing the upper surface **122** of the first dielectric material layer **102** and the upper surface **123** of a conductive via **121** according to various embodiments of the present disclosure. FIG. **9**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **9**A. Referring to FIGS. **9**A and **9**B, an anisotropic etching process may be performed to remove portions of the second dielectric material layer **126** that are not masked by the patterned mask **128** and form an opening **130** extending through the second dielectric material layer **126**. The patterned mask **128** may then be removed by ashing or by dissolution with a solvent. Following the etching process, the upper surface **122** of the first dielectric material layer **102** and the upper surface **123** of a conductive via **121** may be exposed on the bottom of the opening **130**. A side surface **133** of the first conductivity-type semiconductor layer **107** may be exposed along a side of the opening **130**. In some embodiments, the opening **130** in the second dielectric material layer **126** may have a dimension along the second horizontal direction hd**2** that is at least about 5 nm and may be substantially equal to the width dimension of the first conductivity-type semiconductor layer **107** along the second horizontal direction hd**2**. The opening **130** may have a dimension along the first horizontal direction hd**1** of at least about 5 nm.

(108) FIG. **10**A is a top view of the intermediate structure illustrating a first electrode **113** formed within the opening **130** in the second dielectric material layer **126** according to various

embodiments of the present disclosure. FIG. **10**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **10**A. Referring to FIGS. **10**A and **10**B, a conductive material may be deposited within the opening **130** in the second dielectric material layer **126** and over the upper surface **110** of the first conductivity-type semiconductor layer **107** and the upper surface **127** of the second dielectric material layer **126**. In some embodiments, the conductive material may include a metallic liner material and a metallic fill material. The metallic liner material may include a conductive metallic nitride or a conductive metallic carbide such as Ti, Al, TiN, TiN/W, Ti/Al/Ti, TaN, W, Cu, WN, WCN, PdCo, TiC, TaC, and/or WC. The metallic fill material may include W, Cu, Al, Co, Ru, Mo, Ta, Ti, TiN, TaN, alloys thereof, and/or combinations thereof. Other suitable metallic liner and metallic fill materials within the contemplated scope of this disclosure may also be used. The conductive material may be formed by suitable deposition process, which may include one or more of a CVD process, a PVD process, an ALD process, an electroplating process, etc. Other suitable deposition processes are within the contemplated scope of disclosure. Excess portions of the conductive material may be removed from above a horizontal plane including the upper surface **110** of the first conductivity-type semiconductor layer **107** and the upper surface **127** of the second dielectric material layer **126** by a planarization process such as CMP, although other suitable planarization processes may be used. The remaining portion of the conductive material may form the first electrode **113**. The first electrode **113** may contact a conductive via **121** on a bottom surface of the first electrode **113**, and may contact a side surface **133** of the first conductivity-type semiconductor layer **107** on a side surface **133** of the first electrode **113**. A width dimension of the first electrode **113** along the second horizontal direction hd**2** may be substantially equal to the corresponding width dimension of the first conductivity-type semiconductor layer **107**. A height dimension of the first electrode **113** (along a vertical direction) may be substantially equal to the corresponding thickness dimension of the first conductivity-type semiconductor layer **107** (i.e., the upper surface **131** of the first electrode **113** may be substantially coplanar with the upper surface **110** of the first conductivity-type semiconductor layer **107**). A length dimension of the first electrode **113** along the first horizontal direction hd**1** may be at least about 5 nm in various embodiments.

(109) FIG. **11**A is a top view of the intermediate structure illustrating a dielectric isolation layer **111** formed over the first conductivity-type semiconductor layer **107**, the first electrode **113** and the second dielectric material layer **126** according to various embodiments of the present disclosure. FIG. **11**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **11**A. Referring to FIGS. **11**A and **11**B, a dielectric isolation layer **111** may be formed over the upper surface **110** of the first conductivity-type semiconductor layer **107**, the upper surface **131** of the first electrode **113**, and the upper surface **127** of the second dielectric material layer **126**. The dielectric isolation layer **111** may include, but is not limited to, aluminum oxide, silicon oxide, silicon nitride, silicon oxynitride, hafnium oxide, hafnium silicon oxide, hafnium tantalum oxide, hafnium titanium oxide, hafnium zirconium oxide, zirconium oxide, titanium oxide, hafnium dioxide-alumina, or various other insulating structures such as a multi-layer stack structure including alternating insulating layers. The dielectric isolation layer may be deposited by a suitable deposition process (e.g., CVD, ALD, PVD, PECVD, spin coating, etc.) as described above. In some embodiments, a thickness of the dielectric isolation layer **111** may be at least about 5 nm (e.g., between about 5 nm and about 500 nm), although greater and lesser thicknesses may also be utilized.

(110) FIG. **12**A is a top view of the exemplary structure illustrating a continuous second conductivity-type semiconductor layer **109**L formed over the dielectric isolation layer **111** according to various embodiments of the present disclosure. FIG. **12**B is a vertical cross-sectional view of the intermediate structure taken along line A-A′ in FIG. **12**B. Referring to FIGS. **12**A and **12**B, a continuous second conductivity-type semiconductor layer **109**L may be deposited over the upper surface **132** of the dielectric isolation layer **111** via a suitable deposition process, such as

ALD, CVD, PECVD, PVD, etc. A thickness of the continuous second conductivity-type semiconductor layer **109**L may be in a range from approximately 2 nm to approximately 50 nm, although other embodiments may include smaller and larger thicknesses. The continuous second conductivity-type semiconductor layer **109**L may be composed of a suitable semiconductor material, such as a metal oxide semiconductor material.

(111) In some embodiments, the continuous second conductivity-type semiconductor layer **109**L may be an n-type semiconductor material including, but not limited to, one or more of InZnO (IZO), InSnO (indium tin oxide (ITO), In_2O_3, Ga_2O_3, InGaZnO (IGZO), $Al_2O_5Zn_2$, ZnO, InGaO, InWO, ZnO, TiOx, aluminum-doped zinc oxide (AZO), including various combinations and alloys thereof. Other suitable n-type semiconductor materials, such as amorphous silicon, III-V materials, and the like, including combinations (e.g., stacked layers) and/or alloys thereof, may also be utilized. In some embodiments, the continuous second conductivity-type semiconductor layer **109**L may have a composition given by $In_x Ga_y Zn_z MO$, wherein $0<x<1$; $0\leq y\leq 1$; $0\leq z\leq 1$; and M is one of Ti, Al, Ag, Ce, and Sn. In other embodiments, the continuous second conductivity-type semiconductor layer **109**L may include an alloy of oxygen, a group-III element, and a group-V element. Following the deposition of the second conductivity type semiconductor layer **109**L, the exemplary intermediate structure may optionally be annealed. The optional annealing process may be performed at a temperature in a range from 200° C. to 400° C. using a rapid thermal annealing or furnace annealing process. The annealing may be performed in an environment of nitrogen, oxygen, or a mixture thereof.

(112) The continuous second conductivity-type semiconductor layer **109**L may have the opposite conductivity type as the first conductivity-type semiconductor layer **107** described above. Thus, in embodiments in which the first conductivity-type semiconductor layer **107** is composed of a p-type semiconductor material, the continuous second conductivity-type semiconductor layer **109**L may be composed of an n-type semiconductor material. Alternatively, the first conductivity-type semiconductor layer **107** may be composed of an n-type semiconductor material as described above, in which case the continuous second conductivity-type semiconductor layer **109**L may be composed of a suitable p-type semiconductor material as described above with reference to FIGS. **4**A and **4**B.

(113) FIG. **13**A is a top view of the intermediate structure illustrating a patterned mask **135** formed over the continuous second conductivity-type semiconductor layer **109**L according to various embodiments of the present disclosure. FIG. **13**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **13**A. Referring to FIGS. **13**A and **13**B, the patterned mask **135** may be formed by depositing a layer of a photoresist material over the upper surface **134** of the continuous second conductivity-type semiconductor layer **109**L and patterning the photoresist material using lithographic techniques to form a patterned photoresist mask **135** as shown in FIGS. **13**A and **13**B. In some embodiments, the patterned mask **135** may include a strip-shaped portion having a length along the first horizontal direction hd**1** of at least about 15 nm and a width along the second horizontal direction hd**2** of at least about 5 nm. In some embodiments, the patterned mask **135** may have substantially the same length and width dimensions as the underlying first conductivity-type semiconductor layer **107** and may be vertically aligned over the first conductivity-type semiconductor layer **107**. The patterned mask **135** may then be used as a mask to pattern the continuous second conductivity-type semiconductor layer **109**L, as described in greater detail with reference to FIGS. **14**A and **14**B, below.

(114) FIG. **14**A is a top view of the intermediate structure illustrating a discrete second conductivity-type semiconductor layer **109** over the dielectric isolation layer **111** according to various embodiments of the present disclosure. FIG. **14**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **14**A. Referring to FIGS. **14**A and **14**B, an anisotropic etching process may be performed to remove portions of the continuous second conductivity-type semiconductor layer **109**L that are not masked by the patterned mask **135**. The

patterned mask **135** may then be removed by ashing or by dissolution with a solvent. The remaining portion of the continuous first conductivity-type semiconductor layer **109**L may include a discrete first conductivity-type semiconductor layer **109** over the upper surface **132** of the dielectric isolation layer **111**. In some embodiments, the second conductivity-type semiconductor layer **109** may have a length along the first horizontal direction hd**1** of at least about 15 nm and a width along the second horizontal direction hd**2** of at least about 5 nm. In some embodiments, the second conductivity-type semiconductor layer **109** may have substantially the same length and width dimensions as the underlying first conductivity-type semiconductor layer **107** and may be vertically aligned over the first conductivity-type semiconductor layer **107**. In this regard, side surfaces of the second conductivity-type semiconductor layer **109** may be substantially aligned with the corresponding side surfaces of the underlying first conductivity-type semiconductor layer **107**.

(115) FIG. **15**A is a top view of the intermediate structure illustrating a third dielectric material layer **136** formed over the second dielectric material layer **126** and laterally surrounding the second conductivity-type semiconductor layer **109** according to various embodiments of the present disclosure. FIG. **15**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **15**A. Referring to FIGS. **15**A and **15**B, the third dielectric material layer **136** may be deposited over the upper surface **127** of the second dielectric material layer **126** and the upper surface **134** and side surfaces of the second conductivity-type semiconductor layer **109** via a suitable deposition method (e.g., CVD, ALD, PVD, PECVD, spin coating, etc.). The third dielectric material layer **136** may be composed of a suitable dielectric material as described above (e.g., undoped silicate glass, a doped silicate glass, organosilicate glass, silicon oxynitride, silicon carbide nitride, etc.). An optional planarization process, such as a chemical mechanical planarization (CMP) process, may be performed to remove excess portions of the third dielectric material layer **136** from over the upper surface **134** of the second conductivity-type semiconductor layer **109** and provide a planar upper surface **137** of the third dielectric material layer **136**. In the embodiment of FIGS. **15**A and **15**B, the planar upper surface **137** of the third dielectric material layer **136** is substantially coplanar with the upper surface **134** of the second conductivity-type semiconductor layer **109**. In other embodiments, the planar upper surface **137** of the third dielectric material layer **136** may extend over the upper surface **134** of the second conductivity-type semiconductor layer **109**.

(116) FIG. **16**A is a top view of the intermediate structure illustrating a patterned mask **138** formed over the third dielectric material layer **136** and the second conductivity-type semiconductor layer **109** according to various embodiments of the present disclosure. FIG. **16**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **16**A. Referring to FIGS. **16**A and **16**B, the patterned mask **138** may be formed by depositing a layer of a photoresist material over the upper surface **137** of the third dielectric material layer **136** and the upper surface **134** of the second conductivity-type semiconductor layer **109** and patterning the photoresist material using lithographic techniques to form a patterned photoresist mask **138** as shown in FIGS. **16**A and **16**B. The patterned mask **138** may include a strip-shaped opening **139** through the patterned mask **138**, where a portion of the third dielectric material layer **136** may be exposed at the bottom of the opening **139** in the patterned mask **138**. In some embodiments, the opening **139** may have a dimension along the second horizontal direction hd**2** that is at least about 5 nm and may be substantially equal to the width dimension of the second conductivity-type semiconductor layer **109** along the second horizontal direction hd**2**. The opening **139** may have a dimension along the first horizontal direction hd**1** of at least about 5 nm. In some embodiments, the dimensions of the opening **139** along the first and second horizontal directions hd**1** and hd**2** may be substantially identical as the corresponding dimensions of the underlying first electrode **113**, and the opening **129** may be vertically aligned over the first electrode **113**.

(117) FIG. **17**A is a top view of the intermediate structure illustrating an opening **140** formed

through the third dielectric material layer **136** and exposing the upper surface **132** of the dielectric isolation layer **111** according to various embodiments of the present disclosure. FIG. **17**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **17**A. Referring to FIGS. **17**A and **17**B, an anisotropic etching process may be performed to remove portions of the third dielectric material layer **136** that are not masked by the patterned mask **138** and form an opening **140** extending through the third dielectric material layer **136**. The patterned mask **138** may then be removed by ashing or by dissolution with a solvent. Following the etching process, the upper surface **132** of the dielectric isolation layer **111** may be exposed on the bottom of the opening **140**. A side surface **143** of the second conductivity-type semiconductor layer **109** may be exposed along a side of the opening **140**. In some embodiments, the opening **140** in the third dielectric material layer **136** may have a dimension along the second horizontal direction hd**2** that is at least about 5 nm and may be substantially equal to the width dimension of the second conductivity-type semiconductor layer **109** along the second horizontal direction hd**2**. The opening **140** may have a dimension along the first horizontal direction hd**1** of at least about 5 nm. In some embodiments, the dimensions of the opening **140** along the first and second horizontal directions hd**1** and hd**2** may be substantially identical as the corresponding dimensions of the underlying first electrode **113**, and the opening **140** may be vertically aligned over the first electrode **113**.

(118) FIG. **18**A is a top view of the intermediate structure illustrating a second electrode **115** formed within the opening **140** in the third dielectric material layer **136** according to various embodiments of the present disclosure. FIG. **18**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **18**A. Referring to FIGS. **18**A and **18**B, a conductive material may be deposited within the opening **140** in the third dielectric material layer **136** and over the upper surface **134** of the second conductivity-type semiconductor layer **109** and the upper surface **137** of the third dielectric material layer **136**. The conductive material may be composed of one or more suitable conductive materials, such as a metallic liner material and a metallic fill material as described above with reference to the first electrode **113**. The conductive material may be formed by suitable deposition process, such as one or more of a CVD process, a PVD process, an ALD process, an electroplating process, etc. Other suitable deposition processes are within the contemplated scope of disclosure. Excess portions of the conductive material may be removed from above a horizontal plane including the upper surface **134** of the second conductivity-type semiconductor layer **109** and the upper surface **137** of the third dielectric material layer **136** by a planarization process such as CMP, although other suitable planarization processes may be used. The remaining portion of the conductive material may form the second electrode **115**. The second electrode **115** may contact a side surface **143** of the second conductivity-type semiconductor layer **109** on a side surface of the second electrode **115**. A width dimension of the second electrode **115** along the second horizontal direction hd**2** may be substantially equal to the corresponding width dimension of the second conductivity-type semiconductor layer **109**. A height dimension of the second electrode **115** (along a vertical direction) may be substantially equal to the corresponding width dimension of the second conductivity-type semiconductor layer **109** (i.e., the upper surface **141** of the second electrode **115** may be substantially coplanar with the upper surface **134** of the second conductivity-type semiconductor layer **109**). A length dimension of the second electrode **115** along the first horizontal direction hd**1** may be at least about 5 nm in various embodiments. In some embodiments, the dimensions of the second electrode **115** along the first and second horizontal directions hd**1** and hd**2** may be substantially identical as the corresponding dimensions of the underlying first electrode **113**, and the second electrode **115** may be vertically aligned over the first electrode **113**. In this regard, side surfaces of the second electrode **115** may be substantially aligned with the corresponding side surfaces of the underlying first electrode **113**. The dielectric isolation layer **111** may be located between the second electrode **115** and the first electrode **113**.

(119) FIG. **19**A is a top view of the intermediate structure illustrating a patterned mask **144** formed over the third dielectric material layer **136**, the second conductivity-type semiconductor layer **109**,

and the second electrode **115** according to various embodiments of the present disclosure. FIG. **19**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **19**A. Referring to FIGS. **19**A and **19**B, the patterned mask **144** may be formed by depositing a layer of a photoresist material over the upper surface **137** of the third dielectric material layer **136**, the upper surface **134** of the second conductivity-type semiconductor layer **109**, and the upper surface **141** of the second electrode **115** and patterning the photoresist material using lithographic techniques to form a patterned photoresist mask **144** as shown in FIGS. **19**A and **19**B. The patterned mask **144** may include a strip-shaped opening **142** through the patterned mask **144**, where a portion of the third dielectric material layer **136** may be exposed at the bottom of the opening **142** in the patterned mask **144**. In some embodiments, the opening **142** may have a dimension along the second horizontal direction hd**2** that is at least about 5 nm and may be substantially equal to the width dimension(s) of the second conductivity-type semiconductor layer **109** and/or the second electrode **115** along the second horizontal direction hd**2**. The opening **142** may have a dimension along the first horizontal direction hd**1** of at least about 5 nm. The opening **142** may be located adjacent to a side surface of the second conductivity-type semiconductor layer **109** that is opposite to the side surface **143** of the second conductivity-type semiconductor layer **109** that contacts the second electrode **115**.

(120) FIG. **20**A is a top view of the intermediate structure illustrating an opening **148** formed through the third dielectric material layer **136**, the dielectric isolation layer **111**, and the second dielectric material layer **126** and exposing the upper surface **122** of the first dielectric layer **102** according to various embodiments of the present disclosure. FIG. **20**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **20**A. Referring to FIGS. **20**A and **20**B, an anisotropic etching process may be performed to remove portions of the third dielectric material layer **136**, the dielectric isolation layer **111**, and the second dielectric material layer **126** that are not masked by the patterned mask **144** and form an opening **148** extending through the third dielectric material layer **136**, the dielectric isolation layer **111**, and the second dielectric material layer **126**. The patterned mask **144** may then be removed by ashing or by dissolution with a solvent. Following the etching process, the upper surface **122** of the first dielectric material layer **102** and the upper surface **123** of a conductive via **121** may be exposed on the bottom of the opening **148**. A side surface **146** of the second conductivity-type semiconductor layer **109** that is opposite the side surface **143** of the second conductivity-type semiconductor layer **109** that contacts the second electrode **115** may be exposed along a side of the opening **148**. In addition, a side surface **147** of the first conductivity-type semiconductor layer **107** that is opposite the side surface **133** of the first conductivity-type semiconductor layer **109** that contacts the first electrode **113** may also be exposed along a side of the opening **148**. In some embodiments, the opening **148** may have a dimension along the second horizontal direction hd**2** that is at least about 5 nm and may be substantially equal to the width dimension(s) of the second conductivity-type semiconductor layer **109** and/or the first conductivity-type semiconductor layer **107** along the second horizontal direction hd**2**. The opening **148** may have a dimension along the first horizontal direction hd**1** of at least about 5 nm.

(121) FIG. **21**A is a top view of the intermediate structure illustrating a third electrode **117** formed within the opening **148** in the third dielectric material layer **136**, the dielectric isolation layer **111**, and the second dielectric material layer **126** according to various embodiments of the present disclosure. FIG. **21**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **21**B. Referring to FIGS. **21**A and **21**B, a conductive material may be deposited within the opening **148** in the third dielectric material layer **136**, the dielectric isolation layer **111**, and the second dielectric material layer **126**, and over the upper surface **134** of the second conductivity-type semiconductor layer **109**, the upper surface **137** of the third dielectric material layer **136**, and the upper surface **141** of the second electrode **115**. The conductive material may be composed of one or more suitable conductive materials, such as a metallic liner material and a metallic fill

material as described above with reference to the first electrode **113**. The conductive material may be formed by suitable deposition process, such as one or more of a CVD process, a PVD process, an ALD process, an electroplating process, etc. Other suitable deposition processes are within the contemplated scope of disclosure. Excess portions of the conductive material may be removed from above a horizontal plane including the upper surface **134** of the second conductivity-type semiconductor layer **109**, the upper surface **137** of the third dielectric material layer **136**, and the upper surface **141** of the second electrode **115** by a planarization process such as CMP, although other suitable planarization processes may be used. The remaining portion of the conductive material may form the third electrode **117**. The third electrode **117** may contact a conductive via **121** on the bottom surface of the third electrode **117**. A side surface of the third electrode **117** may also contact a side surface **146** of the second conductivity-type semiconductor layer **109** that is opposite the side surface **143** of the second conductivity-type semiconductor layer **109** that contacts the second electrode **115**, and a side surface **147** of the first conductivity-type semiconductor layer **107** that is opposite the side surface **133** of the first conductivity-type semiconductor layer **109** that contacts the first electrode **113**. In some embodiments, the third electrode **117** may have a dimension along the second horizontal direction hd**2** that is at least about 5 nm and may be substantially equal to the width dimension(s) of the second conductivity-type semiconductor layer **109** and/or the first conductivity-type semiconductor layer **107** along the second horizontal direction hd**2**. The third electrode **117** may have a dimension along the first horizontal direction hd**1** of at least about 5 nm. The third electrode **117** may have a height dimension that is equal to the combined height dimensions of the first conductivity-type semiconductor layer **107**, the dielectric isolation layer **111**, and the second conductivity-type semiconductor layer **109** (i.e., an upper surface **145** of the third electrode **117** may be substantially coplanar with the upper surface **134** of the second conductivity-type semiconductor layer **109** and the upper surface **141** of the second electrode **115**).

(122) FIG. **22**A is a top view of the intermediate structure illustrating a patterned mask **149** formed over the third dielectric material layer **136**, the second conductivity-type semiconductor layer **109**, the second electrode **115**, and the third electrode **117** according to various embodiments of the present disclosure. FIG. **22**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **22**A. FIG. **22**C is a vertical cross-section view of the exemplary intermediate structure taken along line B-B′ in FIG. **22**A. Referring to FIGS. **22**A-**22**C, the patterned mask **149** may be formed by depositing a layer of a photoresist material over the upper surface **137** of the third dielectric material layer **136**, the upper surface **134** of the second conductivity-type semiconductor layer **109**, the upper surface **141** of the second electrode **115**, and the upper surface **145** of the third electrode **117** and patterning the photoresist material using lithographic techniques to form a patterned photoresist mask **149** as shown in FIGS. **22**A and **22**B. The patterned mask **149** may include a pair of strip-shaped openings **150** through the patterned mask **149**, where portions of the third dielectric material layer **136** may be exposed at the bottom of each of the openings **150** in the patterned mask **149**. The openings **150** may extend parallel to one another along the first horizontal direction hd**1**. Each of the openings **150** may be located adjacent to a side surface of the second conductivity-type semiconductor layer **109**, as is shown in FIG. **22**C.

(123) FIG. **23**A is a top view of the intermediate structure illustrating a pair of trench openings **151** formed through the third dielectric material layer **136**, the dielectric isolation layer **111**, and the second dielectric material layer **126** and exposing the upper surface **122** of the first dielectric layer **102** according to various embodiments of the present disclosure. FIG. **23**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **23**A. FIG. **23**C is a vertical cross-section view of the intermediate structure taken along line B-B′ in FIG. **23**A. Referring to FIGS. **23**A-**23**C, an anisotropic etching process may be performed to remove portions of the third dielectric material layer **136**, the dielectric isolation layer **111**, and the second dielectric material layer **126** that are not masked by the patterned mask **149** and form a pair of trench openings **151**

extending through the third dielectric material layer **136**, the dielectric isolation layer **111**, and the second dielectric material layer **126**. The patterned mask **149** may then be removed by ashing or by dissolution with a solvent. Following the etching process, the upper surface **122** of the first dielectric material layer **102** may be exposed on the bottom of each of the trench openings **151**. The trench openings **151** may extend parallel to one another along the first horizontal direction hd**1**. Referring to FIG. **23**C, remaining portions of the first conductivity-type semiconductor layer **107**, the dielectric isolation layer **111**, and the second conductivity-type semiconductor layer **109** located between the pair of trench openings **151** may form a layer stack **170** having continuous side surfaces on opposite sides of the layer stack **170** that form outer sidewalls of each of the trench openings **151**. Opposite side surfaces **152***a* and **152***b* of the second conductivity-type semiconductor layer **109** and opposite side surfaces **153***a* and **153***b* of the first conductivity-type semiconductor layer **107** may be exposed along the outer sidewalls of the trench openings **151**.

(124) FIG. **24**A is a top view of the intermediate structure illustrating a gate dielectric layer **103** formed over the third dielectric material layer **136**, the second conductivity-type semiconductor layer **109**, the second electrode **115**, and the third electrode **117** and over the sidewalls and bottom surface of the trench openings **151** according to various embodiments of the present disclosure. FIG. **24**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **24**A. FIG. **24**C is a vertical cross-section view of the intermediate structure taken along line B-B′ in FIG. **24**A. Referring to FIGS. **24**A-**24**C, a gate dielectric layer **103** may be deposited over the upper surface **137** of the third dielectric material layer **136**, the upper surface **134** of the second conductivity-type semiconductor layer **109**, the upper surface **141** of the second electrode **115** and the upper surface **145** of the third electrode **117**, and along the sidewalls and bottom surfaces of each of the trench openings **151**. The gate dielectric layer **103** may include a suitable dielectric material, such as, for example, one or more of silicon oxide, silicon nitride, silicon oxynitride, hafnium oxide, hafnium silicon oxide, tantalum oxide, hafnium tantalum oxide, hafnium titanium oxide, hafnium zirconium oxide, zirconium oxide, titanium oxide, aluminum oxide, hafnium dioxide-alumina, or various other insulating structures such as a multi-layer stack structure including alternating insulating layers. Other suitable dielectric materials are within the contemplated scope of disclosure. In various embodiments, the gate dielectric layer **103** may be formed by a suitable conformal deposition process such as ALD, CVD, PECVD, PVD, etc. A thickness of the gate dielectric layer **103** may be in a range from approximately 2 nm to approximately 20 nm, although other embodiments may include smaller and larger thicknesses. Following the deposition of the gate dielectric layer **103**, the intermediate structure may optionally be annealed. The optional annealing process may be performed at a temperature in a range from 200° C. to 400° C. using a rapid thermal annealing or furnace annealing process. The annealing may be performed in an environment of nitrogen, oxygen, or a mixture thereof.

(125) FIG. **25**A is a top view of the intermediate structure illustrating a dielectric fill layer **154** formed over the gate dielectric layer **103** and filling a remaining volume of the trench openings **151** and a patterned mask **155** formed over the dielectric fill layer **154** according to various embodiments of the present disclosure. FIG. **25**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **25**A. FIG. **25**C is a vertical cross-section view of the exemplary intermediate structure taken along line B-B′ in FIG. **25**A. Referring to FIGS. **25**A-**25**C, a dielectric fill layer **154** may be deposited over the upper surface of the gate dielectric layer **103** via a suitable deposition method (e.g., CVD, ALD, PVD, PECVD, spin coating, etc.). The dielectric fill layer **154** may fill the remaining volume of the pair of trench openings **151**. The dielectric fill layer **154** may be composed of a suitable dielectric material as described above. An optional planarization process, such as a chemical mechanical planarization (CMP) process, may be performed to provide a planar upper surface of the dielectric fill layer **154**.

(126) A patterned mask **155** may be formed over the dielectric fill layer **154** by depositing a layer of a photoresist material over the upper surface of the dielectric fill layer **154** and patterning the

photoresist material using lithographic techniques to form a patterned photoresist mask **155** as shown in FIGS. **25**A-**25**C. The patterned mask **155** may be similar to the patterned mask **149** described above with reference to FIGS. **22**A-**22**C and may include a pair of strip-shaped openings **156** extending parallel to one another along the first horizontal direction hd**1**. The dielectric fill layer **154** may be exposed at the bottom of each of the openings **156** in the patterned mask **155**. In some embodiments, a length dimension of each of the openings **156** along the first horizontal direction hd**1** may be at least about 10 nm. In some embodiments, the length dimension of the openings **156** may be substantially commensurate with a distance between the third electrode **117** and the second electrode **115** along the first horizontal direction hd**1**. In various embodiments, a width dimension of each of the openings **156** along the second horizontal direction hd**2** may be at least about 10 nm. As shown in FIG. **25**C, the patterned mask **155** may overlie a horizontally-extending **103***c* portion of the gate dielectric layer **103** located over the upper surface **134** of the second conductivity-type semiconductor layer **109** and vertically-extending portions **103***a* and **103***b* of the gate dielectric layer **103** that contact side surfaces **152***a* and **152***b* of the second conductivity-type semiconductor layer **109** and side surfaces **153***a* and **153***b* of the first conductivity-type semiconductor layer **107**.
(127) FIG. **26**A is a top view of the intermediate structure illustrating a pair of trench openings **157** formed through the dielectric fill layer **154** and portions of the gate dielectric layer **103** and exposing the upper surface **122** of the first dielectric layer **102** according to various embodiments of the present disclosure. FIG. **26**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **26**A. FIG. **26**C is a vertical cross-section view of the intermediate structure taken along line B-B′ in FIG. **26**A. Referring to FIGS. **26**A-**26**C, an anisotropic etching process may be performed to remove portions of the dielectric fill layer **154** and the gate dielectric layer **103**, and optionally portions of the third dielectric layer **136**, the dielectric isolation layer **111**, and the second dielectric layer **126** to provide a pair of trench openings **157** extending parallel to one another along the first horizontal direction hd**1**. The patterned mask **155** may then be removed by ashing or by dissolution with a solvent. Following the etching process, the upper surface **122** of the first dielectric layer **102** may be exposed at the bottom of each of the trench openings **157**. As shown in FIG. **26**C, a portion of the gate dielectric layer **103** may surround the layer stack **170** over two side surfaces and an upper surface of the layer stack **170**. That is, a first vertically-extending portion **103***a* of the gate dielectric layer **103** may extend over and contact a first side surface **153***a* of the first conductivity-type semiconductor layer **107**, a first side surface of the dielectric isolation layer **111**, and a first side surface **152***a* of the second conductivity-type semiconductor layer **109**. A second vertically-extending portion **103***b* of the gate dielectric layer **103** may extend over and contact a second side surface **153***b* of the first conductivity-type semiconductor layer **107**, a second side surface of the dielectric isolation layer **111**, and a second side surface **152***b* of the second conductivity-type semiconductor layer **109**. A horizontally-extending portion **103***c* of the gate dielectric layer **103** may extend over and contact the upper surface **134** of the second conductivity-type semiconductor layer **109** between the first vertically-extending portion **103***a* and the second vertical-extending portion **103***b* of the dielectric layer **103**. The first vertically-extending portion **103***a* and the second vertical-extending portion **103***b* of the gate dielectric layer **103** may each form an outer sidewall of a respective trench opening **157**.
(128) FIG. **27**A is a top view of a semiconductor device structure **100** including a pair of gate electrodes **105***a* and **105***b* extending on opposite sides of a layer stack **170** according to various embodiments of the present disclosure. FIG. **27**B is a vertical cross-section view of the intermediate structure taken along line A-A′ in FIG. **27**A. FIG. **28**C is a vertical cross-section view of the intermediate structure taken along line B-B′ in FIG. **27**A. FIG. **27**D is a vertical cross-section view of the intermediate structure taken along line C-C′ in FIG. **27**A. FIG. **27**E is a vertical cross-section view of the intermediate structure taken along line D-D′ in FIG. **27**A. Referring to FIGS. **26**A-**26**E, a conductive material may be deposited within the trench openings **157** and over the

upper surface of the dielectric fill layer **154**. The conductive material may fill the trench openings **157** to at least the level of a horizontal plane including the upper surface of the horizontally-extending portion **103***c* of the gate dielectric layer **103**. A planarization process, such as a CMP process and/or an etching process, may be used to remove portions of the conductive material and the dielectric fill layer **154** from above the horizontal plane including the upper surface of the horizontally-extending portion **103***c* of the gate dielectric layer **103** and provide discrete gate electrodes **105***a* and **105***b* located within each of the trench openings **157**. A first gate electrode **105***a* may contact the first vertically-extending portion **103***a* of the gate dielectric layer **103** over a first side surface of the layer stack **170**, and a second gate electrode **105***b* may contact the second vertically-extending portion **103***b* of the gate dielectric layer **103** over a second side surface of the layer stack **170**.

(129) The first gate electrode **105***a* and the second gate electrode **105***b* may be composed of a suitable conductive material as described above, such as TiN, W, WN, WCN, Co, PdCo, Mo, Cu, TaN, Ti, Al, etc., including combinations and alloys thereof. Other suitable electrically conductive materials are within the contemplated scope of disclosure. In some embodiments, the first gate electrode **105***a* and the second gate electrode **105***b* may include a metallic liner material and a metallic fill material as described above. The first gate electrode **105***a* and the second gate electrode **105***b* may be formed by suitable deposition process, which may include one or more of a CVD process, a PVD process, an ALD process, an electroplating process, etc. Other suitable deposition processes are within the contemplated scope of disclosure. In various embodiments, each of the gate electrodes **105***a* and **105***b* may have a length dimension along the first horizontal direction hd**1** of at least about 10 nm, and a width dimension along the second horizontal direction hd**2** of at least about 10 nm. An upper surface **158** of each of the gate electrodes **105***a* and **105***b* may be substantially coplanar with the upper surface of the gate dielectric layer **103**.

(130) Referring to FIGS. **27**A-**27**E, a semiconductor device structure **100** according to an embodiment of the present invention is illustrated. The semiconductor device structure includes a layer stack **170** including a first conductivity-type semiconductor layer **107**, a dielectric isolation layer **111** over the first conductivity-type semiconductor layer **107**, and a second conductivity-type semiconductor layer **109** over the dielectric isolation layer **111**. A first electrode **113** may be located on a first end **101** of the semiconductor device structure **100** and may contact a first side surface **133** of the first conductivity-type semiconductor layer **107**. A second electrode **115** may be located on the first end **101** of the semiconductor device structure **100** and may contact a first side surface **143** of the second conductivity-type semiconductor layer **109**. The dielectric isolation layer **111** may extend between the first electrode **113** and the second electrode **115**. A third electrode **117** may be located on a second end **104** of the semiconductor device structure **100** that is opposite the first end **101** and may contact a second side surface **147** of the first conductivity-type semiconductor layer **107**, a side surface of the dielectric isolation layer **111**, and a second side surface **146** of the second conductivity-type semiconductor layer **109**. A pair of gate electrodes **105***a* may extend along opposite side surfaces of the layer stack **170** and a gate dielectric layer **103** may extend between each of the gate electrodes **105***a* and **105***b* and each of the respective side surfaces of the layer stack **170**.

(131) In various embodiments, the semiconductor device structure **100** may be configured as a NOT gate or inverter circuit such that the application of a given input signal having a first logic state at the first gate electrode **105***a* and the second gate electrode **105***b* will produce an output signal from the third electrode **117** having the complementary logic state. In this regard, a high voltage input signal at the first gate electrode **105***a* and the second gate electrode **105***b* results in a low voltage output signal at the third electrode **117**, and a low voltage input signal at the first gate electrode **105***a* and the second gate electrode **105***b* results in a high voltage output signal at the third electrode **117**. In various embodiments, the semiconductor device structure **100** may be fabricated using a BEOL process using thin-film metal-oxide semiconductor materials.

Accordingly, this may enable computing operations to be performed, at least in part, within one or more interconnect levels of a semiconductor integrated circuit (IC) die, which may enable improved integration density and/or increased processing power of the semiconductor IC die.
(132) FIG. **28**A is a top view of a semiconductor device structure **100** including a fourth dielectric material layer **159** formed over the gate dielectric layer **103** and the gate electrodes **105***a* and **105***b* and a plurality of conductive vias **160***a* and **160***b* located within the fourth dielectric material layer **159** according to various embodiments of the present disclosure. FIG. **28**B is a vertical cross-section view of the semiconductor device structure **100** of taken along line A-A′ in FIG. **28**A. FIG. **28**C is a vertical cross-section view of the semiconductor device structure **100** of taken along line B-B′ in FIG. **28**A. Referring to FIGS. **28**A-**28**C, the fourth dielectric material layer **159** may be deposited over the upper surface of the gate dielectric layer **103** and the upper surfaces **158** of the first and second gate electrodes **105***a* and **105***b* via a suitable deposition method (e.g., CVD, ALD, PVD, PECVD, spin coating, etc.). The fourth dielectric material layer **159** may be composed of a suitable dielectric material as described above (e.g., undoped silicate glass, a doped silicate glass, organosilicate glass, silicon oxynitride, silicon carbide nitride, etc.). Metal interconnect features, including metal lines **161**, **163** and conductive vias **160***a*, **160***b*, **162**, may be formed within the first dielectric material layer **102** (e.g., via a single or dual damascene process). (The metal lines **161** and **163** are not shown in FIG. **28**A for purposes of clarity). A via **162** may electrically connect a metal line **163** to the second electrode **115** as shown in FIG. **28**B. A pair of vias **160***a* and **160***b* may electrically connect a metal line **161** to each of the gate electrodes **105***a* and **105***b*, as shown in FIG. **28**C.
(133) FIG. **29** is a flow diagram illustrating a method **300** for fabricating a semiconductor device structure **100** according to various embodiments of the present disclosure. Referring to FIGS. **4**A-**6**B and **29**, in step **301** of embodiment method **300** a first conductivity-type semiconductor layer **107** may be formed. Referring to FIGS. **7**A-**10**B and **29**, in step **303** of embodiment method **300**, a first electrode **113** may be formed contacting a first side surface **133** of the first conductivity-type semiconductor layer **107**. Referring to FIGS. **11**A-B and **29**, in step **305** of embodiment method **300**, a dielectric isolation layer **111** may be formed over the first conductivity-type semiconductor layer **107** and the first electrode **113**. Referring to FIGS. **12**A-**14**B and **29**, in step **307** of embodiment method **300**, a second conductivity-type semiconductor layer **109** may be formed over the dielectric isolation layer **111**. Referring to FIGS. **15**A-**18**B and **29**, in step **309** of embodiment method **300**, a second electrode **115** may be formed contacting a first side surface **143** of the second conductivity-type semiconductor layer **109**. Referring to FIGS. **19**A-**21**B and **29**, in step **311** of embodiment method **300**, a third electrode **117** may be formed contacting a second side surface **147** of the first conductivity-type semiconductor layer **107**, a side surface of the dielectric isolation layer **111**, and a second side surface **146** of the second conductivity-type semiconductor layer **109**. Referring to FIGS. **22**A-**26**C and **29**, in step **313** of embodiment method **300**, a gate dielectric layer **103** may be formed over opposite third and fourth side surfaces **153***a* and **153***b* of the first conductivity-type semiconductor layer **107**, opposite side surfaces of the dielectric isolation layer **111**, and opposite third and fourth side surfaces **152***a* and **152***b* of the second conductivity-type semiconductor layer **109**. Referring to FIGS. **27**A-**27**E and **29**, in step **315** of embodiment method **300**, a first gate electrode **105***a* may be formed contacting the gate dielectric layer **103** over the third side surface **153***a* of the first conductivity-type semiconductor layer **107** and the third side surface **152***a* of the second conductivity-type semiconductor layer **109**, and a second gate electrode **105***b* may be formed contacting the gate dielectric layer **103** over the fourth side surface **153***b* of the first conductivity-type semiconductor layer **107** and the fourth side surface **152***b* of the second conductivity-type semiconductor layer **109**.
(134) Referring to all drawings and according to various embodiments of the present disclosure, a semiconductor device structure **100** is provided. The semiconductor device structure **100** may include a layer stack **170** having a first conductivity-type semiconductor layer **107**, a dielectric

isolation layer **111** over the first conductivity-type semiconductor layer **107**, and a second conductivity-type semiconductor layer **109** over the dielectric isolation layer **111**. A first electrode **113** may be adjacent to a first end **101** of the layer stack **170** and may contact a first side surface **133** of the first conductivity-type semiconductor layer **107**. A second electrode **115** may be adjacent to the first end **101** of the layer stack and may contact a first side surface **143** of the second conductivity-type semiconductor layer **109**. A third electrode **117** may be adjacent to a second end **104** of the layer stack **170** opposite the first end **101** of the layer stack **170** and may contact a second side surface **147** of the first conductivity-type semiconductor layer **107**, a side surface of the dielectric isolation layer **111**, and a second side surface **146** of the second conductivity-type semiconductor layer **109**. A gate dielectric layer **103** may contact opposite side surfaces of the layer stack **170** over at least a portion of a length of the layer stack **170** between the first end and the second end of the layer stack **170**. A pair of gate electrodes **105***a* and **105***b* may contact the gate dielectric layer **103** and extend over the opposite side surfaces of the layer stack **170** over at least a portion of the length of the layer stack **170** between the first end and the second end of the layer stack **170**.
(135) In an embodiment, at least one of the first conductivity-type semiconductor layer **107** or the second conductivity-type semiconductor layer **109** include metal-oxide semiconductors.
(136) In another embodiment, the first conductivity-type semiconductor layer **107** includes a p-type semiconductor layer, and the second conductivity-type semiconductor layer **109** includes an n-type semiconductor layer.
(137) In another embodiment, the p-type semiconductor layer includes one or more of NiO, SnO, and Cu_2O.
(138) In another embodiment, the n-type semiconductor layer includes one or more of amorphous silicon, $Al_2O_5Zn_2$ doped ZnO, InGaZnO, InGaO, InWO, InZnO, InSnO, Ga_2O_3, ZnO, GaO, Ga_2O_3, InO, In_2O_3, InZnO, ZnO, TiOx, and a III-V semiconductor material.
(139) In another embodiment, a length of the layer stack **170** along a first horizontal direction hd**1** between the first end **101** and the second end **104** of the layer stack **170** is 15 nm or more.
(140) In another embodiment, a width of each of the first electrode **113**, the second electrode **115**, and the third electrode **117** along a second horizontal direction hd**2** that is perpendicular to the first horizontal direction hd**1** is equal to a width of the layer stack **170** between the opposite side surfaces of the layer stack **170**.
(141) In another embodiment, a thickness of the first conductivity-type semiconductor layer **107** and the second conductivity-type semiconductor layer **109** is between 2 nm and 50 nm, a thickness of the dielectric isolation layer **111** is at least 5 nm, and a thickness of the gate dielectric layer **103** is between 2 nm and 20 nm.
(142) In another embodiment, the gate dielectric layer **103** includes a first portion **103***a* that extends over a first side surface of the layer stack **170**, a second portion **103***b* that extends over a second side surface of the layer stack **170**, and a third portion **103***c* that extends over an upper surface **134** of the second conductivity-type semiconductor layer **109** between the first portion **103***a* and the second portion **103***b* of the gate dielectric layer **103**.
(143) In another embodiment, the first conductivity-type semiconductor layer **103** is formed over an interconnect-level dielectric layer **102** having a horizontal surface **122**, and wherein the interfacing surfaces of the first conductivity-type semiconductor layer **107**, the dielectric isolation layer **111**, and the second conductivity-type semiconductor layer **109** are substantially parallel to the horizontal surface **122** of the interconnect-level dielectric layer **102**.
(144) In another embodiment, the interconnect-level dielectric layer **102** further includes one or more electrical interconnect structures (**120**, **121**), and wherein the first electrode **113** and the third electrode **117** are electrically connected to the one or more electrical interconnect structures (**120**, **121**).

(145) In another embodiment, a dielectric material matrix (**126**, **111**, **136**, **103**, **159**) is located over and laterally surrounds the semiconductor device structure **100**, the dielectric material matrix (**126**, **111**, **136**, **103**, **159**) including one or more electrical interconnect structures (**160***a*, **160***b*, **161**, **162**, **163**), and the second electrode **115** and the pair of gate electrodes **105***a* and **105***b* are electrically connected to the one or more electrical interconnect structures (**160***a*, **160***b*, **161**, **162**, **163**) in the dielectric material matrix (**126**, **111**, **136**, **103**, **159**).
(146) In another embodiment, the first electrode **113** is electrically connected to a voltage supply terminal VDD, the second electrode **115** is electrically connected to a ground terminal GND, the pair of gate electrodes **105***a* and **105***b* are electrically connected to a signal input terminal Vin, and the third electrode **117** is electrically connected to a signal output terminal Vout.
(147) Another embodiment is drawn to a semiconductor device including a substrate **8** having a semiconductor material layer **9**; a plurality of device structures **700** formed on or in the semiconductor material layer **9**, at least one interconnect-level dielectric layer (**601**, **610**, **620**, **630**, **640**, **102**) over the substrate **8** and the device structures **700** and including conductive interconnect structures (**612**, **618**, **622**, **628**, **632**, **638**, **642**, **648**, **120**, **121**), and a NOT gate logic circuit **100** located over an interconnect-level dielectric layer (**601**, **610**, **620**, **630**, **640**, **102**). The NOT gate logic circuit **100** may include a first conductivity-type semiconductor layer **107**, a second conductivity-type semiconductor layer **109**, a first electrode **113** contacting a first side surface **133** of the first conductivity-type semiconductor layer **107**, a second electrode **115** contacting a first side surface **143** of the second conductivity-type semiconductor layer **109**, a dielectric isolation layer **111** extending between the first conductivity-type semiconductor layer **107** and the second conductivity-type semiconductor layer **109** and between the first electrode **113** and the second electrode **115**, a third electrode **117** contacting a second side surface **147** of the first conductivity-type semiconductor layer **107**, a second side surface **146** of the second conductivity-type semiconductor layer **109**, and a side surface of the dielectric isolation layer **111**, a gate dielectric layer **103** contacting the first conductivity-type semiconductor layer **107** and the second conductivity-type semiconductor layer **109**, and a gate electrode (**105***a*, **105***b*) contacting the gate dielectric layer **103**.
(148) In an embodiment, interfacing surfaces of the first conductivity-type semiconductor layer **107**, the dielectric isolation layer **111**, and the second conductivity-type semiconductor layer **109** are parallel to a major surface of the substrate **8**.
(149) In another embodiment, the first conductivity-type semiconductor layer **107** includes a p-type semiconductor layer, the second conductivity-type semiconductor layer **109** includes an n-type semiconductor layer, the first electrode **113** is electrically connected to a voltage source terminal VDD, the second electrode **115** is electrically connected to a ground terminal GND, the gate electrode (**105***a*, **105***b*) is electrically connected to an input signal terminal Vin, and the third electrode **117** is electrically connected to an output signal terminal Vout.
(150) Another embodiment is drawn to a method of fabricating a semiconductor device structure that includes forming a first conductivity-type semiconductor layer **107**, forming a first electrode **113** contacting a first side surface **133** of the first conductivity-type semiconductor layer **107**, forming a dielectric isolation layer **111** over the first conductivity-type semiconductor layer **107** and the first electrode **113**, forming a second conductivity-type semiconductor layer **109** over the dielectric isolation layer **111**, forming a second electrode **115** contacting a first side surface **143** of the second conductivity-type semiconductor layer **109**, forming a third electrode **117** contacting a second side surface **147** of the first conductivity-type semiconductor layer **107**, a side surface of the dielectric isolation layer **111**, and a second side surface **146** of the second conductivity-type semiconductor layer **109**, forming a gate dielectric layer **103** over opposite third and fourth side surfaces **153***a* and **153***b* of the first conductivity-type semiconductor layer **107**, opposite side surfaces of the dielectric isolation layer **111**, and opposite third and fourth side surfaces **152***a* and **152***b* of the second conductivity-type semiconductor layer **109**, and forming a first gate electrode

105*a* contacting the gate dielectric layer **103** over the third side surface **153***a* of the first conductivity-type semiconductor layer **107** and the third side surface **152***a* of the second conductivity-type semiconductor layer **109**, and a second gate electrode **105***b* contacting the gate dielectric layer **103** over the fourth side surface **153***b* of the first conductivity-type semiconductor layer **107** and the fourth side surface **152***b* of the second conductivity-type semiconductor layer **109**.

(151) In an embodiment, forming the gate dielectric layer **103** includes forming a pair of trench openings **151**, wherein the opposite third and fourth side surfaces **152***a* and **152***b* of the second conductivity-type semiconductor layer **109**, the opposite side surfaces of the dielectric isolation layer **111**, and the opposite third and fourth side surfaces **153***a* and **153***b* of the first conductivity-type semiconductor layer **107** are exposed along sidewalls of the respective trench openings; and depositing the gate dielectric layer **103** over an upper surface **134** of the second conductivity-type semiconductor layer **109** and along the sidewalls and bottom surfaces of each of the trench openings **151**.

(152) In another embodiment, forming the first gate electrode **105***a* and the second gate electrode **105***b* includes filling the pair of trench openings **151** with a dielectric fill layer **154**, forming a pair of second trench openings **157**, wherein the gate dielectric layer **103** is exposed along sidewalls of the respective second trench openings **157**, and depositing an electrically conductive material within each of the second trench openings **157** and contacting the gate dielectric layer **103** to provide the first gate electrode **105***a* and the second gate electrode **105***b*.

(153) In another embodiment, forming a plurality of first conductive vias **121** in a first dielectric material layer **102**, wherein the first electrode **113** and the third electrode **117** are formed over the first dielectric layer **102** such that the first electrode **113** and the third electrode **117** each contact a first conductive via **121**, forming a second dielectric material layer **159** over gate dielectric layer **103** and the first and second gate electrodes **105***a* and **105***b*, and forming a plurality of second conductive vias (**160***a*, **160***b*, **162**) within the second dielectric material layer **159** and contacting the second electrode **115**, the first gate electrode **105***a* and the second gate electrode **105***b*.

(154) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of this disclosure. Those skilled in the art should appreciate that they may readily use this disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of this disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of this disclosure.

Claims

1. A semiconductor device structure, comprising: a layer stack comprising: a first conductivity-type semiconductor layer; a dielectric isolation layer over the first conductivity-type semiconductor layer; and a second conductivity-type semiconductor layer over the dielectric isolation layer; a first electrode adjacent to a first end of the layer stack and contacting a first side surface of the first conductivity-type semiconductor layer; a second electrode adjacent to the first end of the layer stack and contacting a first side surface of the second conductivity-type semiconductor layer; a third electrode adjacent to a second end of the layer stack opposite the first end of the layer stack and contacting a second side surface of the first conductivity-type semiconductor layer, a side surface of the dielectric isolation layer, and a second side surface of the second conductivity-type semiconductor layer; a gate dielectric layer contacting opposite side surfaces of the layer stack over at least a portion of a length of the layer stack between the first end and the second end of the layer stack; and a pair of gate electrodes contacting the gate dielectric layer and extending over opposite side surfaces of the layer stack over at least a portion of the length of the layer stack between the

first end and the second end of the layer stack.
2. The semiconductor device structure of claim 1, wherein at least one of the first conductivity-type semiconductor layer or the second conductivity-type semiconductor layer comprise metal-oxide semiconductors.
3. The semiconductor device structure of claim 1, wherein the gate dielectric layer includes a first portion that extends over a first side surface of the layer stack, a second portion that extends over a second side surface of the layer stack, and a third portion that extends over an upper surface of the second conductivity-type semiconductor layer between the first portion and the second portion of the gate dielectric layer.
4. The semiconductor device structure of claim 1, wherein the first electrode is electrically connected to a voltage supply terminal, the second electrode is electrically connected to a ground terminal, the pair of gate electrodes are electrically connected to a signal input terminal, and the third electrode is electrically connected to a signal output terminal.
5. The semiconductor device structure of claim 1, wherein the first conductivity-type semiconductor layer comprises a p-type semiconductor layer, and the second conductivity-type semiconductor layer comprises an n-type semiconductor layer.
6. The semiconductor device structure of claim 5, wherein the p-type semiconductor layer comprises one or more of NiO, SnO, and Cu.sub.2O.
7. The semiconductor device structure of claim 5, wherein the n-type semiconductor layer comprises one or more of amorphous silicon, Al.sub.2O.sub.5Zn.sub.2 doped ZnO, InGaZnO, InGaO, InWO, InZnO, InSnO, Ga.sub.2O.sub.3, ZnO, GaO, Ga.sub.2O.sub.3, InO, In.sub.2O.sub.3, InZnO, ZnO, TiOx, and a III-V semiconductor material.
8. The semiconductor device structure of claim 1, wherein a length of the layer stack along a first horizontal direction between the first end and the second end of the layer stack is at least 15 nm.
9. The semiconductor device structure of claim 8, wherein a width of each of the first electrode, the second electrode, and the third electrode along a second horizontal direction that is perpendicular to the first horizontal direction is equal to a width of the layer stack between the opposite side surfaces of the layer stack.
10. The semiconductor device structure of claim 8, wherein a thickness of the first conductivity-type semiconductor layer and the second conductivity-type semiconductor layer is between 2 nm and 50 nm, a thickness of the dielectric isolation layer is at least 5 nm, and a thickness of the gate dielectric layer is between 2 nm and 20 nm.
11. The semiconductor device structure of claim 1, wherein the first conductivity-type semiconductor layer is formed over an interconnect-level dielectric layer having a horizontal surface, and wherein interfacing surfaces of the first conductivity-type semiconductor layer, the dielectric isolation layer, and the second conductivity-type semiconductor layer are substantially parallel to the horizontal surface of the interconnect-level dielectric layer.
12. The semiconductor device structure of claim 11, wherein the interconnect-level dielectric layer further comprises one or more electrical interconnect structures, and wherein the first electrode and the third electrode are electrically connected to the one or more electrical interconnect structures.
13. The semiconductor device structure of claim 12, wherein a dielectric material matrix is located over and laterally surrounds the semiconductor device structure, the dielectric material matrix comprising one or more electrical interconnect structures, and the second electrode and the pair of gate electrodes are electrically connected to the one or more electrical interconnect structures in the dielectric material matrix.
14. A semiconductor device, comprising: a substrate comprising a semiconductor material layer; a plurality of device structures formed on or in the semiconductor material layer; at least one interconnect-level dielectric layer over the substrate and the plurality of device structures and comprising conductive interconnect structures; and a NOT gate logic circuit located over an interconnect-level dielectric layer, the NOT gate logic circuit comprising: a first conductivity-type

semiconductor layer; a second conductivity-type semiconductor layer; a first electrode contacting a first side surface of the first conductivity-type semiconductor layer; a second electrode contacting a first side surface of the second conductivity-type semiconductor layer; a dielectric isolation layer extending between the first conductivity-type semiconductor layer and the second conductivity-type semiconductor layer and between the first electrode and the second electrode; a third electrode contacting a second side surface of the first conductivity-type semiconductor layer, a second side surface of the second conductivity-type semiconductor layer, and a side surface of the dielectric isolation layer; a gate dielectric layer contacting the first conductivity-type semiconductor layer and the second conductivity-type semiconductor layer; and a gate electrode contacting the gate dielectric layer.

15. The semiconductor device of claim 14, wherein interfacing surfaces of the first conductivity-type semiconductor layer, the dielectric isolation layer, and the second conductivity-type semiconductor layer are substantially parallel to a major surface of the substrate.

16. The semiconductor device of claim 14, wherein the first conductivity-type semiconductor layer comprises a p-type semiconductor layer, the second conductivity-type semiconductor layer comprises an n-type semiconductor layer, the first electrode is electrically connected to a voltage source terminal, the second electrode is electrically connected to a ground terminal, the gate electrode is electrically connected to an input signal terminal, and the third electrode is electrically connected to an output signal terminal.

17. A method of fabricating a semiconductor device structure, comprising: forming a first conductivity-type semiconductor layer; forming a first electrode contacting a first side surface of the first conductivity-type semiconductor layer; forming a dielectric isolation layer over the first conductivity-type semiconductor layer and the first electrode; forming a second conductivity-type semiconductor layer over the dielectric isolation layer; forming a second electrode contacting a first side surface of the second conductivity-type semiconductor layer; forming a third electrode contacting a second side surface of the first conductivity-type semiconductor layer, a side surface of the dielectric isolation layer, and a second side surface of the second conductivity-type semiconductor layer; forming a gate dielectric layer over opposite third and fourth side surfaces of the second conductivity-type semiconductor layer, opposite side surfaces of the dielectric isolation layer, and opposite third and fourth side surfaces of the first conductivity-type semiconductor layer; and forming a first gate electrode contacting the gate dielectric layer over the third side surface of the first conductivity-type semiconductor layer and the third side surface of the second conductivity-type semiconductor layer, and a second gate electrode contacting the gate dielectric layer over the fourth side surface of the first conductivity-type semiconductor layer and the fourth side surface of the second conductivity-type semiconductor layer.

18. The method of claim 17, wherein forming the gate dielectric layer comprises: forming a pair of trench openings, wherein the opposite third and fourth side surfaces of the second conductivity-type semiconductor layer, the opposite side surfaces of the dielectric isolation layer, and the opposite third and fourth side surfaces of the first conductivity-type semiconductor layer are exposed along sidewalls of the respective trench openings; and depositing the gate dielectric layer over an upper surface of the second conductivity-type semiconductor layer and along the sidewalls and bottom surfaces of each of the trench openings.

19. The method of claim 18, wherein forming the first gate electrode and the second gate electrode comprises: filling the pair of trench openings with a dielectric fill layer; forming a pair of second trench openings, wherein the gate dielectric layer is exposed along sidewalls of the respective second trench openings; and depositing an electrically conductive material within each of the second trench openings and contacting the gate dielectric layer to provide the first gate electrode and the second gate electrode.

20. The method of claim 18, further comprising: forming a plurality of first conductive vias in a first dielectric material layer, wherein the first electrode and the third electrode are formed over the

first dielectric layer such that the first electrode and the third electrode each contact a first conductive via; forming a second dielectric material layer over gate dielectric layer and the first and second gate electrodes; and forming a plurality of second conductive vias within the second dielectric material layer and contacting the second electrode, the first gate electrode and the second gate electrode.